Problems from *Discrete Mathematics* *Elementary and Beyond* *by László Lovász et.al. [4]*

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Contents

0.1 Preface 8

1 Let's Count! 11

1.1 A Party 11

1.2 Sets and the Like 15

1.3 The Number of Subsets 21

1.4 The Approximate Number of Subsets 24

1.5 Sequences 24

1.6 Permutations 29

1.7 The Number of 'Ordered' k-subsets of an n-set 31

1.8 The Number of k-subsets of an n-set 34

2 Combinatorial Tools 51

2.1 Induction 51

2.2 Comparing and Estimating Numbers 58

2.3 Inclusion-Exclusion 60

These solutions are mainly for my idea of fun, but are also dedicated to all the combinatorists past and present.

Box 1: How I have Used A.I. in This Work

Update on 27 September 2026: I have used ChatGPT to draw a few tikz pictures in this work.

I can't believe I had an urge to write this declaration. A number of developments in the 'outside world' dominated currently by the achievements of A.I. prompted me to do this. As I read this book and solved problems from it, I decided to write down how I have used A.I. (by that I mean Large Language Models, A.I. coding assistants, etc.) in this work. I will make this a practice. This policy is maintained throughout this work. The current A.I. models appear as L.L.M's first. Therefore, A.I. and L.L.M. mean the same thing in the context of this work.

1. By default, I have not discussed the exposition or problems with an L.L.M. If a particular topic is discussed with an L.L.M, it is mentioned. If there's no mention, I guarantee that no 'conversation with L.L.M' is involved.
2. All programs are written without a coding assistant, unless stated otherwise explicitly. Minimum editing, code-completion facilities of an I.D.E. are used.
3. Solutions to all the problems in this book (unlike, for example, Knuth's books) are known. There is no reason for me to consult an L.L.M, except when the given exposition and/or solutions are unclear or incorrect. Again, such occurrences are expected to be few, if any.
4. I do not hate A.I. It's just that, for whatever reason, I don't feel the need to solicit its help in understanding known material lucidly presented in this book.

Reflection 1 (Introduction)**.** These are the author's solutions to the problems from this amazing textbook ([4]). They may occasionally sound like author's conversation with the reader (or even himself). Even if they are personal and appear in the public domain, they may help the reader, although the author isn't exactly sure how. He wrote them for 0) the love of mathematics and writing, 1) LaTeX practice (of writing hopefully readable mathematics, albeit longer than appearing in standard texts), and 2) reliable archival (paper, on which most work is created, tends to get lost) that incurs minimal overhead.

These notes are interspersed with 'reflection boxes' like this one. They are meant to express author's 'rough' thinking process (mental catharsis of sorts), of which frustration is often an inseparable part. The author writes mostly in first person for an authentic conversational tone.

My best friend, Deepa Joshi, once asked me to write about the 'trail of events' that led to whatever I was reading at the time she observed that I had opened yet another 'good book.' I try to fulfill her wish: Reading Don Knuth's books is what I normally do[a]. While reading one of his books, it occurred to me that fixing some big gaps in my understanding of discrete mathematics (maybe I have forgotten things) would be a step in the right direction. I chose two books, from a wondrous garden of excellent books on combinatorics[b]: 1) Lovász's present book ([4]), which is supposedly elementary (at least partly), and 2) Robert Beeler's book, *How to Count*.

So, I am starting with solutions to the problems from this book. I may occasionally write about the exposition, but, unlike my other attempts, these notes are primarily my solutions to the problems occurring in the book.

Why Lovász? Because he is one[c] of Knuth's 'heroes of combinatorics' ([5]). Another reason was that I had access to the digital copy of this book[d].

[a] His books are like mathematical axioms; I don't need a 'trail of events' that lead me to his books.

[b] The choice is very, very difficult.

[c] One other is the great Richard Stanley; I need to work hard to enjoy his book, Enumerative Combinatorics, which I plan to defer reading for now.

[d] Thanks to Springer, who made it available legally to the public during the recent pandemic years.

0.1 Preface

Audience, level, and treatment–a description of such matters is what prefaces are supposed to be about.

— Paul Halmos
I Want to be a Mathematician [1]

The aim of this book is not to cover 'discrete mathematics' in depth (it should be clear from the description above that such a task would be ill-defined and impossible anyway). Rather, we discuss a number of selected results and methods, mostly from the areas of combinatorics and graph theory, with a little elementary number theory, probability, and combinatorial geometry.

...

Another important ingredient of mathematics is problem solving. You won't be able to learn any mathematics without dirtying your hands and trying out the ideas you learn about in the solution of problems. To some, this may sound frightening, but in fact, most people pursue this type of activity almost every day: Everybody who plays a game of chess or solves a puzzle is solving discrete mathematical problems. The reader is strongly advised to answer the questions posed in the text and to go through the problems at the end of each chapter of this book. Treat it as puzzle solving, and if you find that some idea that you came up with in the solution plays some role later, be satisfied that you are beginning to get the essence of how mathematics develops.

— Lovász et.al.
PREFACE TO [4]

Chapter 1

Let's Count!

1.1 A Party

This section introduces the art of counting gently, using a running example of seven friends partying at Alice's house on her birthday: Alice, Bob, Carl, Diane, Eve, Frank, and George. They initially sit at a round table and eat.

Counting problems can be daunting.

My main challenges in solving counting problems:

- Interpret the given problem correctly. This makes a combinatorial problem as a 'word problem.' The language skill is critical.
- Think about a given counting problem in more than one equivalent way.

Authors choose not to explicitly declare the multiplication principle which is useful many a time:

Definition 1 (Multiplication Principle)**.** If one choice can be made in m_1 ways and another choice in m_2 ways, and the two choices are independent, then the two choices together can be made in $m_1 \times m_2$ ways.

In general, if there are n choices, the first one can be made in m_1 ways, second in m_2 ways, and so on, and the choices are independent, then the total number of ways in which those choices can be made together is $m_1 \times m_2 \times \cdots \times m_n$.

Exercise 1.1.1. *How many ways can these people be seated at the table if Alice, too, can sit anywhere?*

Solution.

Since Alice can sit at one of the seven chairs, she can make the choice in seven ways. Once Alice makes a choice, Bob can choose any one of the remaining six seats. Is Bob's choice *independent* of Alice's? Clearly, Bob can't choose the seat that Alice chose. However, regardless of that, Bob has six seats to choose from. The number of Bob's choices is independent of the choice Alice makes. From multiplication principle it follows that Alice and Bob can be seated in $7 \times 6 = 42$ different ways.

This process continues until George has the Hobson's choice of taking the only remaining seat when it's his turn to choose. It follows from the multiplication principle (See definition [1]) that the total number of ways these 7 friends can *choose* 7 seats together is $7 \times 6 \times 5 \times \cdots \times 2 \times 1 = 7! = 5040$.

We have counted the number of ways a seating arrangement can be done, given the constraints, without listing the arrangements and enumerating them. Enumerating is one way of counting, but it is not the only way. Counting *without* enumerating is a huge time-saver, provided our argument (based on the multiplication principle in this case) is correct.

On the other hand, *enumerating* such arrangements is also an important aspect of combinatorial problems.

Pairing Chess Players

6 friends ($2n$, in general) want to play chess on 3 chessboards (n, in general). In how many ways can they be paired? Consider that the chessboards are numbers $1, 2, 3, \ldots, n$.

Well, we must convey the constraints clearly. The book calls a set of these constraints 'a sense of pairing' and names them after the player who suggests them. At least three pairing schemes are suggested:

1. Bob's sense ("I think you should not count it as a different pairing if two people at the same board switch places," says Bob, "and it shouldn't matter which pair sits at which board.")
2. Carl's sense ("Yes, I think we have to agree on what the question really means," adds Carl. "If we include in it who plays white on each board, then if a pair switches places we do get a different matching. But Bob is right that it doesn't matter which pair uses which board.")
3. Diane's sense: ("What do you mean it does not matter? You sit at the first board, which is closest to the peanuts, and I sit at the last, which is farthest," says Diane.)

Exercise 1.1.2. *What is the number of pairings in Carl's sense (when it matters who sits on which side of the board, but the boards are all alike), and in Diane's sense (when it is the other way around)?*

Solution.

Box 2: Wrong Answers Only!

The book calculates the number of pairings in Bob's sense: 15. Indeed, as a game of chess is played between two opponents, the number of pairings is the same as the number of games played when any two players are chosen (with 6 choices for the first player, and having made that choice, 5 for the second) and they play only one game between them, although two are possible, for example, George (White) versus Carol (Black) and Carol (White) versus George (Black). The number of games is:

$$\frac{6 \times 5}{2} = 15$$

In Carl's sense, this number doubles. The Diane-Frank pairing is distinct from the Frank-Diane pairing. We always give the first player White pieces and the second player Black. Thus, every pair plays two games and there are 15 pairs as above. Therefore, the number of pairings in Carl's sense is 30.

In Diane's sense, we consider the Bob-Carol pairing on board 1 distinct from the Bob-Carol pairing on board 2 or board 3. It's like pairing them up on board 1 first (15 pairings, since sides don't matter), then on board 2 (15 more pairings), and lastly on board 3 (still 15 more pairings) giving 45 pairings.

Reflection 2 (I want to work harder)**.** My initial answer above (box [1.1]) shows my gross misunderstanding of the problem. I had read the relevant section from the book and thought I had understood it, but alas, I hadn't! A serious discussion with my son, Apoorv Mhaswade, revealed it to me. I am very grateful to him!

This also means that passive reading is not enough. For serious study of known material, one must actively engage in quizzing oneself. One must experiment, solve problems, and, if solutions are provided, check them thoroughly. I thank Lovász for providing the correct solutions. While studying by oneself, this can feel onerous, but there's no choice to get more familiar with a vast body of known technical matter. For new research, this means becoming more immune to confirmation bias.

Combinatorial problems act as if they have a knack of fooling you! Wrong thinking about a concrete problem can lead to the correct answer, giving you a false confidence that you are right. My attempt above demonstrates that I completely misinterpreted the problem and each of the 'senses' in which the counting is specified. This is frustrating, but it emphasizes the indispensability of hard work.

First, we need to pair 6 players across 3 chessboards. This is different from 6 people shaking hands. The

handshake custom doesn't care on which side of the handshake you are and, unlike the chessboard, there's no 'handshake board.'

Alice initially thinks that the number of ways of pairing six people on three chessboards is equivalent to the number of ways of seating them on six chairs and placing a chessboard between each pair across it. If this is how want to pair them up, then let that be. The number of ways of seating six friends in six chairs simultaneously is clearly $6! = 720$. Alice is right in her own right although each player may be paired with the same opponent on the same board.

It's important to note (although not mentioned explicitly) that all the players sit and start playing their 'round' simultaneously. It's *not* that players finish playing on board 1 (while board 2 and board 3 are unused), then board 2, and finally board 3; in other words, they *do not use the available boards sequentially.*

The arrangement initially proposed by Alice can also be realized another way: For the board 1, we can choose the first player in 6 ways and the second player in 5 ways, since sides matter (Bob with White pieces versus Carl with Black must be counted differently from Bob with Black versus Carl with White). Those are $6 \times 5 = 30$ pairings. Once those two players occupy board 1, we only have $6 - 2 = 4$ choices for the first player on board 2 and 3 for the second. That gives $4 \times 3 = 12$ independent choices for board 2. Similarly, $2 \times 1 = 2$ independent choices for board 3. Together (from Multiplication Principle (definition [1])), the total number of choices for the matching or pairing is

$$30 \times 12 \times 2 = 720$$

Listing all the pairs across three boards is mechanical.

However, Bob thinks that once two are paired, they must not be paired again[a]. This means that the board 1 pair can only be chosen 5 ways[b] (you choose a player and their opponent can be chosen from the remaining 5, giving 5 pairs); simultaneously, the board 2 pair can only be independently chosen 3 ways; and the board 3 pair can be chosen independently once.

Therefore, the total number of choices for pairing in Bob's sense is

$$5 \times 3 \times 1 = 15$$

This number of pairings is the same in any scenario. One scenario is listed below (table [6]).

Board 1	Board 2	Board 3
Bob-Carl	Diane-Eve	Frank-George
Bob-Diane	Carl-Frank	Eve-George
Bob-Eve	Carl-George	Diane-Frank
Bob-Frank	Carl-Eve	Diane-George
Bob-George	Carl-Diane	Eve-Frank

Table 1.1: A Scenario for Pairings in Bob's Sense (Bob *Sticks* to Board 1)

In this scenario, Bob only plays on board 1. It's possible that Carl plays only on board 1, or Diane only plays on board 1, and so on. In each of those scenarios (there are 6), the *number of pairings stays the same*, 15.

Carl's wording ("If we include in it who plays white…") is arguably confusing. What I now think he means is that a Bob (White)-Carl (Black) pair on a given board is distinct from Bob (Black)-Carl (White) pair *on the same board*, but not distinct on another board: A pair plays twice on a board; they don't play again on any other board.

Thus, Carl wants to count each pair on the same board twice. Therefore, the total number of choices for pairing in Carl's sense is

$$10 \times 6 \times 2 = 120$$

, or equivalently,

$$(5 \times 2) \times (3 \times 2) \times (1 \times 2) = 5 \times 3 \times 1 \times 2^3 = 15 \times 2^3 = 120$$

Carl multiplies Bob's count by 2^n.

Diane's assertion is also tricky to state clearly. I think she wants to retain Bob's sense of not pairing the same players again on the same board, but not retain his sense of not pairing them on another board. Consider an arrangement of three boards in Bob's sense: Board 1: Bob-Carl, Board 2: Diane-Eve, Board 3: Frank-George. This 3-board arrangement is counted by Bob. However, Diane counts this and all its rearrangements (permutations) distinctly:

Board 1	Board 2	Board 3
Bob-Carl	Diane-Eve	Frank-George
Bob-Carl	Frank-George	Diane-Eve
Diane-Eve	Bob-Carl	Frank-George
Diane-Eve	Frank-George	Bob-Carl
Frank-George	Bob-Carl	Diane-Eve
Frank-George	Diane-Eve	Bob-Carl

Table 1.2: Diane Counts What Bob and Carl Don't

These are the placements of 3 pairs in 3 places (across each chessboard). Their number is $3 \times 2 \times 1 = 3! = 6$. Each such 3-pair from Bob's pairings is multiplied by this number by Diane. Therefore, in Diane's sense, the number of pairings is $15 \times 3! = 15 \times 6 = 90$.

Diane multiplies Bob's count by $n!$.

It turns out that Alice's initial estimate, $(2n)!$, equals Bob's count $((2n-1) \times (2n-3) \times \cdots \times 3 \times 1)$ multiplied by Carl's factor (2^n) and Diane's factor $(n!)$.

[a] To emphasize, they must not be paired again on any board.
[b] **Not** $6 \times 5 = 30$ as in Alice's scheme, where each pair plays many times on the same board.

Exercise 1.1.3. *What is the number of pairings (in all the various senses as above) in a party of* 10*?*

Solution.

Box 3: Wrong Answers Again!

In Bob's sense,

$$\frac{10 \times 9}{2} = 45$$

In Carl's sense, $10 \times 9 = 90$.

In Diane's sense, $45 \times 3 = 135$.

Reflection 3. Answers in the above box are also wrong because I did that exercise before I reflected on the previous one (See reflection [2]).

The correct answers follow.

In Bob's sense, $9 \times 7 \times 5 \times 3 \times 1 = 945$.

In Carl's sense, $945 \times 2 \times 2 \times 2 \times 2 \times 2 = 945 \times 2^5 = 30{,}440$.

In Diane's sense, $945 \times 5 \times 4 \times 3 \times 2 \times 1 = 945 \times 5! = 113{,}400$.

Interestingly,

$$945 \times 2^5 \times 5! = 10!$$

1.2 Sets and the Like

Sets are complicated, but here's an introduction. Sets are a fundamental tool that helps with counting problems.

Definition 2 (Set)**.** Any collection of distinct objects, called elements, is a set.

Definition 3 (Empty Set or Null Set)**.** A set with no elements is called an Empty Set. It's denoted by $\emptyset$.

Definition 4 (Cardinality of a Set)**.** The number of elements of a set is its cardinality.

For example, the cardinality $|\mathbb{P}|$ of the set $\mathbb{P} = \{1, 4, 9, 16\}$ is 5.

We'll not discuss infinite sets (sets whose cardinality is ∞) here.

Definition 5 (Subset)**.** A set $\mathbb{A}$ is called a subset of set $\mathbb{B}$ if every element of $\mathbb{A}$ is an element of $\mathbb{B}$.

In other words, $\mathbb{A}$ consists of certain elements of $\mathbb{B}$.

We can allow $\mathbb{A}$ to consist of all elements of $\mathbb{B}$ (in which case $\mathbb{A} = \mathbb{B}$) or none of them (in which case $\mathbb{A} = \emptyset$), and still consider it a subset. So the empty set is a subset of every set.

This is denoted by $\mathbb{A} \subseteq \mathbb{B}$. If $|\mathbb{A}| \neq |\mathbb{B}|$, then $\mathbb{A}$ is called a *proper subset* of $\mathbb{B}$, denoted by $\mathbb{A} \subset \mathbb{B}$.

We consider operations on two sets.

Definition 6 (Intersection)**.** The intersection, $\mathbb{A} \cap \mathbb{B}$, of two sets, $\mathbb{A}$ and $\mathbb{B}$, is the set consisting of elements common to both the sets.

When there are no common elements, this set is $\emptyset$, and the two sets are said to be *disjoint* with each other.

Definition 7 (Union)**.** The union, $\mathbb{A} \cup \mathbb{B}$, of two sets, $\mathbb{A}$ and $\mathbb{B}$, is the set consisting of elements from either set.

Definition 8 (Difference)**.** The difference, $\mathbb{A} \setminus \mathbb{B}$, between two sets, $\mathbb{A}$ and $\mathbb{B}$, is the set consisting of elements from $\mathbb{A}$ but not $\mathbb{B}$. The difference, $\mathbb{B} \setminus \mathbb{A}$, between two sets, $\mathbb{B}$ and $\mathbb{A}$, is the set consisting of elements from $\mathbb{B}$ but not $\mathbb{A}$.

Definition 9 (Symmetric Difference)**.** The symmetric difference, $\mathbb{A} \triangle \mathbb{B}$, between two sets, $\mathbb{A}$ and $\mathbb{B}$, is the set consisting of elements from exactly one of the two sets.

The exposition here brightly illuminates the relevant portion of set theory! The authors compare sets with natural numbers and draw parallels between them to the extent possible. They challenge readers to think of set intersection like multiplication (this may not be natural), set union like addition (quite natural), and the two set differences like subtraction (somewhat natural) of numbers. These similarities and differences beget some remarkable set identities that may reveal as a positive side-effect of solving the following exercises.

Exercise 1.2.1. *Name sets whose elements are (a) buildings, (b) people, (c) students, (d) trees, (e) numbers, (f) points*

Solution.

(a) Apartment complex.

(b) Organization.

(c) School magazine committee.

(d) Orchard.

(e) Days in a leap year's months.

(f) Circle.

Exercise 1.2.2. *What are the elements of the following sets: (a) army, (b) mankind, (c) library, (d) the animal kingdom?*

Solution. It's not easy to list the *type* of every element of the given sets.

(a) Soldiers.

(b) Humans.

(c) Books.

(d) Animals.

Exercise 1.2.3. *Name sets having cardinality (a)* 52*, (b)* 13*, (c)* 32*, (d)* 100*, (e)* 90*, (f)* 2,000,000.

Solution.

Reflection 4. I spent time figuring out some realistic, but mathematically interesting sets of given cardinalities. A time-consuming, but interesting exercise! Which set has exactly 2 million elements? A city of that population–that's alright, but is it interesting?

(a) $\mathbb{D} = \{$♠-1, ♠-2, …, ♠-K, ♠-A, ♡-1, …, ♢-9, …, ♣-A$\}$: Deck of playing cards. $|\mathbb{D}| = 52$.

(b) $\mathbb{H}$: Hand of bridge (the card game). $|\mathbb{H}| = 13$.

(c) $\mathbb{B}_5 = \{$00000, 00001, …, 01011, …, 11110, 11111$\}$: Binary 5-strings. $|\mathbb{B}_5| = 32$. Also, $\mathcal{P}(\mathbb{S}) = \{\emptyset, \{1\}, \{4\}, \ldots, \{1, 4, 9, 16, 25\}\}$: Power-set of the set $\mathbb{S} = \{1, 4, 9, 16, 25\}$. $|\mathcal{P}(\mathbb{S})| = 32$.

(d) $\mathbb{O}$: Possible 'breakfast-lunch-dinner' orders at an all-day restaurant with 5 choices for breakfast, 5 for lunch, and 4 for dinner. $|\mathbb{O}| = 100$.

(e) $\mathbb{D}_{15} = \{$AC,AD,AE,...,BD,BE,...,MO$\}$: Diagonals of a regular 15-gon, ABCDEFGHIJKLMNO. $|\mathbb{D}_{15}| = 90$.

(f) $\mathfrak{L}$: Black-on-white or white-on-black car license plates of the form <c><c><c><d><d><d>, where '<c>' is any letter from 'A' to 'J' (both inclusive) and '<d>' is any decimal digit. An element of this set is EGA340; another is EGA340.

$\mathfrak{L} = \{$AAA000, AAA001, …, JJJ998, JJJ999, AAA000, AAA001, …, JJJ998, JJJ999$\}$. $|\mathfrak{L}| = 2{,}000{,}000$.

Reflection 5. Below are the textbook answers to this exercise. I argue that mine are better ;-) (a) all cards in a deck; (b) all spades in a deck; (c) a deck of Swiss cards[a]; (d) nonnegative integers with at most two digits; (e) nonnegative integers with exactly two digits; (f) inhabitants of Budapest, Hungary.

[a] I guess they mean piquet cards.

Exercise 1.2.4. *What are the elements of the following (admittedly peculiar) set:* $\{Alice, \{1\}\}$

Solution. Alice (a person) and a (singular) set whose only element is 1.

Exercise 1.2.5. *Is an 'element of a set' a special case of a 'subset of a set'?*

Solution. I don't think so. An element of a set is a single object, which might not be a set. A subset of a set, however, is always a set. How can we allow 'a special case' of a set to not be a set?

Reflection 6. The textbook answer simply says, "No!"

Exercise 1.2.6. *List all subsets of* $\mathbb{S} = \{0, 1, 3\}$. *How many do you get?*

Solution. $\emptyset, \{0\}, \{1\}, \{3\}, \{0, 1\}, \{0, 3\}, \{1, 3\}, \{0, 1, 3\}$. Eight.

Exercise 1.2.7. *Define at least three sets of which* $\mathbb{A} = \{$*Alice, Diane, Eve*$\}$ *is a subset.*

Solution.

Left as an exercise to the readers of the note-taking readers of Lovász.

Exercise 1.2.8. *List all subsets of* $\mathbb{S} = \{a, b, c, d, e\}$, *containing* a *but not* b.

Solution.

These are the 8 subsets of the set $\{c, d, e\}$, to each of which we add an a.

Reflection 7 (Always review your answers)**.** I got this right while checking my answers after solving all the exercises, but before looking at the answers provided at the end of the book. I wrote "These are the 16 subsets of the set $\{a, c, d, e\}$" before!

Exercise 1.2.9. *Define a set of which both* $\{1, 3, 4\}$ *and* $\{0, 3, 5\}$ *are subsets. Find such a set with the smallest possible number of elements.*

Solution. $\mathbb{S} = \{0, 1, 2, 3, 4, 5\}$. However, this is not such a set with smallest number of elements; $\mathbb{T} = \{0, 1, 3, 4, 5\}$ is.

Exercise 1.2.10. *(a) Which set would you call the union of* $\mathbb{A} = \{a, b, c\}$, $\mathbb{B} = \{a, b, d\}$, *and* $\mathbb{C} = \{b, c, d, e\}$*?*

(b) Find the union of the first two sets, and then the union of this with the third. Also, find the union of the last two sets, and then the union of this with the first set. Try to formulate what you have observed.

(c) Give a definition of the union of more than two sets.

Solution.

(a) The set $\mathbb{U} = \{a, b, c, d, e\}$ can be called the union of *three* given sets, denoted by $\mathbb{A} \cup \mathbb{B} \cup \mathbb{C}$.

(b) $\mathbb{A} \cup \mathbb{B} = \{a, b, c, d\}$.
$(\mathbb{A} \cup \mathbb{B}) \cup \mathbb{C} = \{a, b, c, d, e\}$
$\mathbb{B} \cup \mathbb{C} = \{a, b, c, d, e\}$.
$(\mathbb{B} \cup \mathbb{C}) \cup \mathbb{A} = \{a, b, c, d, e\}$.
Conjecture: $(\mathbb{A} \cup \mathbb{B}) \cup \mathbb{C} = \mathbb{A} \cup (\mathbb{B} \cup \mathbb{C})$. This is already proved (proof tbd) for equation [1.3] in the book.

(c) This definition assumes the definition [7] of union of two sets.

Definition 10 ((Proposed) Union of $n \geq 2$ Sets)**.** Consider n sets, $\mathbb{A}_1, \mathbb{A}_2, \ldots, \mathbb{A}_n$. $\mathbb{U}$, their union, is defined as the set you get after you carry out the following algorithm:

1: $\mathsf{U} \leftarrow \mathsf{A}_1 \cup \mathsf{A}_2$
2: $\mathsf{i} \leftarrow 3$
3: **while** $i \leq n$ **do** ▷ Take the union of two sets $n - 2$ times more
4: $\quad \mathsf{U} \leftarrow \mathsf{U} \cup \mathsf{A}_i$
5: $\quad i \leftarrow i + 1$
6: **end while**
7: ▷ U is the union of *all* the given sets

Reflection 8. The textbook defines it more succinctly and correctly. I like their definition better:

Definition 11 (Union of sets)**.** The union of any set of sets consists of those elements that are elements of at least one of the sets.

Exercise 1.2.11. *Explain the connection between the notion of the union of sets and exercise [1.2.9].*

Solution. The union, $\mathbb{U}$, of given sets is the set with smallest cardinality, such that every given set is a subset of $\mathbb{U}$.

Exercise 1.2.12. *We form the union of a set $\mathbb{A}$ with 5 elements and a set $\mathbb{B}$ with 9 elements. Which of the following numbers can we get as the cardinality of their union $\mathbb{U}$: 4, 6, 9, 10, 14, 20?*

Solution. $\mathbb{U}$ must have at least 9 elements because $\mathbb{B}$, which has 9 elements, is its subset. Therefore, it couldn't have 4 or 6 elements. Similarly, it couldn't have 20 elements; to get to that many elements, after we choose all the elements of $\mathbb{B}$, we have only $\mathbb{A}$ to choose from, which can only offer at most 5 elements. Thus, $\mathbb{U}$ could have any of $9, 10, 11, 12, 13$, or 14 elements. In this case, 9 (all the elements of $\mathbb{A}$ are also in $\mathbb{B}$; $\mathbb{A} \subset \mathbb{B}$), 10 ($\mathbb{A}$ and $\mathbb{B}$ have four common elements) or 14 ($\mathbb{A}$ and $\mathbb{B}$ are disjoint).

Reflection 9 (A rare typo)**.** The textbook gives $6, 9, 10, 14$ as the answer. Including 6 is a typo.

Exercise 1.2.13. *We form the union of two sets. We know that one of them has n elements and the other has m elements. What can we infer about the cardinality of their union?*

Solution. Two extremities as far as union $\mathbb{U}$ of two sets is concerned are 1) one of them is a subset of another, and 2) the two sets are disjoint. In 1), one of the sets (the subset) does not contribute new elements to the union. Therefore, in that case, the set with higher cardinality determines that of the union. In 2), since all elements from both the sets must be in the union, the union has $m + n$ elements; the union of any two sets cannot have a cardinality larger than that. $\therefore \max(m, n) \leq |\mathbb{U}| \leq (m + n)$.

Exercise 1.2.14. *What is the intersection of*

(a) The sets $\mathbb{A} = \{0, 1, 3\}$ and $\mathbb{B} = \{1, 2, 3\}$,

(b) The set of girls in this class and the set of boys in this class, and

(c) The set of prime numbers and the set of even numbers?

Solution.

(a) $\mathbb{A} \cap \mathbb{B} = \{1, 3\}$.

(b) $\emptyset$[1]

(c) $\{2\}$.

Exercise 1.2.15. *We form the intersection of two sets. We know that one of them, $\mathbb{A}$, has n elements and the other, $\mathbb{B}$, has m elements. What can we infer about the cardinality of their intersection?*

Solution. If $\mathbb{A}, \mathbb{B}$ are *disjoint*, their intersection, $\mathbb{A} \cap \mathbb{B}$ is the empty set, $\emptyset$. The cardinality of the intersection is incremented for each common element. The number of common elements in any two sets can at most equal the number of elements in the set with a smaller cardinality. $\therefore 0 \leq |\mathbb{A} \cap \mathbb{B}| \leq \min(|\mathbb{A}|, |\mathbb{B}|)$.

Exercise 1.2.16. *Prove*

[1] We are sensitive to both the difficulty of one's sex-assignment at birth and we are sensitive to one's gender identity, but to keep things simple, we stick to the 'binary notions' of human body.

(a) *The commutativity of the union and intersection of any two sets,* $\mathbb{A}, \mathbb{B}$*:*

$$\begin{aligned} \mathbb{A} \cap \mathbb{B} &= \mathbb{B} \cap \mathbb{A} \\ \mathbb{A} \cup \mathbb{B} &= \mathbb{B} \cup \mathbb{A} \end{aligned} \tag{1.1}$$

(b) *The associativity of the union and intersection of any three sets,* $\mathbb{A}, \mathbb{B}, \mathbb{C}$*:*

$$\begin{aligned} (\mathbb{A} \cap \mathbb{B}) \cap \mathbb{C} &= \mathbb{A} \cap (\mathbb{B} \cap \mathbb{C}) \\ (\mathbb{A} \cup \mathbb{B}) \cup \mathbb{C} &= \mathbb{A} \cup (\mathbb{B} \cup \mathbb{C}) \end{aligned} \tag{1.2}$$

(c) *The set-specific property, distribution of union over intersection involving any three sets,* $\mathbb{A}, \mathbb{B}, \mathbb{C}$*:*

$$\mathbb{A} \cup (\mathbb{B} \cap \mathbb{C}) = (\mathbb{A} \cup \mathbb{B}) \cap (\mathbb{A} \cup \mathbb{C}) \tag{1.3}$$

Solution.

(a) *Proof.* By definition, $\mathbb{A} \cup \mathbb{B}$ contains elements in either $\mathbb{A}$ or $\mathbb{B}$.

By definition, $\mathbb{B} \cup \mathbb{A}$ contains elements in either $\mathbb{B}$ or $\mathbb{A}$.

Since the 'elements in either $\mathbb{B}$ or $\mathbb{A}$' are the same as the 'elements in either $\mathbb{A}$ or $\mathbb{B}$,' the two sets, $\mathbb{A} \cup \mathbb{B}$ and $\mathbb{B} \cup \mathbb{A}$, are the same.

By definition, $\mathbb{A} \cap \mathbb{B}$ contains elements in both $\mathbb{A}$ and $\mathbb{B}$.

By definition, $\mathbb{B} \cap \mathbb{A}$ contains elements in both $\mathbb{B}$ and $\mathbb{A}$.

Since the 'elements in both $\mathbb{B}$ and $\mathbb{A}$' are the same as the 'elements in both $\mathbb{A}$ and $\mathbb{B}$,' the two sets, $\mathbb{A} \cap \mathbb{B}$ and $\mathbb{B} \cap \mathbb{A}$, are the same. ■

(b) *Proof.* Items common to all the sets are the same as the items common to first two and the last or the items common to the first and the last two. ■

(c) *Proof.* We can take a 'truth-table-like' approach and show exhaustively that 1) every element that belongs to the set $\mathbb{A} \cup (\mathbb{B} \cap \mathbb{C})$ also belongs to the set $(\mathbb{A} \cup \mathbb{B}) \cap (\mathbb{A} \cup \mathbb{C})$, and 2) every element that does not belong to the set $\mathbb{A} \cup (\mathbb{B} \cap \mathbb{C})$ also does not belong to the set set $(\mathbb{A} \cup \mathbb{B}) \cap (\mathbb{A} \cup \mathbb{C})$. ■

Exercise 1.2.17. *Prove that, for any two sets* $\mathbb{A}, \mathbb{B}$*,*

$$|\mathbb{A} \cup \mathbb{B}| + |\mathbb{A} \cap \mathbb{B}| = |\mathbb{A}| + |\mathbb{B}| \tag{1.4}$$

Solution.

> **Reflection 10.** A lot of math is practiced with pen on paper; we are fortunate to have both! I was stumped by this at first, but somehow referred to two earlier exercises and made a connection that showed a ray of hope. I think the proof is right. Let's see...

Proof. We reproduce the inequalities from the solutions to exercises [1.2.13] and [1.2.15].

$$\begin{aligned} \max(|\mathbb{A}|, |\mathbb{B}|) \leq |\mathbb{A} \cup \mathbb{B}| &\leq (|\mathbb{A}| + |\mathbb{B}|) \\ 0 \leq |\mathbb{A} \cap \mathbb{B}| &\leq \min(|\mathbb{A}|, |\mathbb{B}|) \end{aligned}$$

Adding these two inequalities, we get

$$|\mathbb{A} \cup \mathbb{B}| + |\mathbb{A} \cap \mathbb{B}| \leq (|\mathbb{A}| + |\mathbb{B}|) + \min(|\mathbb{A}|, |\mathbb{B}|)$$

Subtracting the nonnegative number, $|\mathbb{A}| + |\mathbb{B}|$, from both sides of the inequality gives us the following inequality:

$$|\mathbb{A} \cup \mathbb{B}| + |\mathbb{A} \cap \mathbb{B}| - (|\mathbb{A}| + |\mathbb{B}|) \leq \min(|\mathbb{A}|, |\mathbb{B}|)$$

The only value of $|\mathbb{A} \cup \mathbb{B}| + |\mathbb{A} \cap \mathbb{B}| - (|\mathbb{A}| + |\mathbb{B}|)$ that satisfies the above inequality is 0, because, the minimum of the cardinalities of two sets is 0; there's no set with a negative cardinality.

$$\therefore |\mathbb{A} \cup \mathbb{B}| + |\mathbb{A} \cap \mathbb{B}| - (|\mathbb{A}| + |\mathbb{B}|) = 0,$$

giving

$$|\mathbb{A} \cup \mathbb{B}| + |\mathbb{A} \cap \mathbb{B}| = (|\mathbb{A}| + |\mathbb{B}|)$$

■

Reflection 11 (A succinct proof)**.** The common elements of $\mathbb{A}$ and $\mathbb{B}$ are counted twice on both sides; the elements in either $\mathbb{A}$ or $\mathbb{B}$ but not both are counted once on both sides.

(I really like this one; I should've thought of it. Indeed, brevity is the soul of wit.)

Exercise 1.2.18. *(a) What is the symmetric difference (See definition [9]) of the set*[2] $\mathbb{Z}_{\geq 0}$ *of nonnegative integers and the set* $\mathbb{E}$ *of even integers (*$\mathbb{E} = \{\dots, -4, -2, 0, 2, 4, \dots\}$ *contains both negative and positive even integers).*

(b) Form the symmetric difference of $\mathbb{A}$ *and* $\mathbb{B}$ *to get a set* $\mathbb{C}$*. Form the symmetric difference of* $\mathbb{A}$ *and* $\mathbb{C}$*. What did you get? Give a proof of the answer.*

Solution.

(a)

$$\mathbb{Z}_{\geq 0} = \{0, 1, 2, 3, 4, 5, \dots\}$$
$$\mathbb{E} = \{\dots, -4, -2, 0, 2, 4, \dots\}$$

Positive odd numbers belong *only* to $\mathbb{Z}_{\geq 0}$, and negative even numbers *only* to $\mathbb{E}$.

$$\therefore \mathbb{Z}_{\geq 0} \triangle \mathbb{E} = \{\dots, -6, -4, -2, 1, 3, 5, \dots\}$$

(b) $\mathbb{B}$.

Proof. We make an exclusive argument based on elements of the sets, $\mathbb{A}, \mathbb{B}, \mathbb{C}$. We know that

$$\mathbb{C} = \mathbb{A} \triangle \mathbb{B}$$

For every element $e \in \mathbb{C}$, by definition, either 1) $e \in \mathbb{A}, e \notin \mathbb{B}$, or 2) $e \in \mathbb{B}, e \notin \mathbb{A}$.

Now, when we form $\mathbb{A} \triangle \mathbb{C}$, again, by definition, we must not choose those elements of $\mathbb{C}$ that are also in $\mathbb{A}$. Thus, we'll end up choosing the elements of $\mathbb{C}$ that are only members of $\mathbb{B}$ (and not of $\mathbb{A}$). Therefore, $(\mathbb{A} \triangle \mathbb{B}) \triangle \mathbb{A} = \mathbb{B}$. ■

Reflection 12. The textbook does not provide a proof. I believe mine is valid.

[2] We prefer this notation to $\mathbb{Z}_+$ that the book uses; see `https://math.stackexchange.com/a/1402225/83821`.

1.3 The Number of Subsets

Theorem 1.3.1. *A set with n elements has 2^n subsets.*

This section is beautifully written. It provides two proofs of this theorem. It also carefully distinguishes between choosing a subset and enumerating all subsets of a given set.

I wish, however, that they had more clearly stated the idea of 'ordering' a set.

Exercise 1.3.1. *Under the correspondence between numbers and subsets described above,*

(a) Which numbers correspond to subsets with 1 element?

(b) Which number corresponds to the whole set?

(c) Which sets correspond to even numbers?

Solution.

(a) In the binary representations of the subsets with 1 element we must have exactly one 1. Such representations, owing to the place-value system, correspond to *positive* powers of 2.

(b) In the binary representation of the subset that is the same as the given set, we must have n 1's, which corresponds to $2^n - 1$.

(c) The rightmost bit in binary representations of even numbers is 0. This means the *last element* in the preferred order of set elements is not in the corresponding subsets.

Exercise 1.3.2. *What is the number of subsets of a set with n elements, containing a given element?*

Solution. Presence of a 'given set element' in any subset is indicated by a 1 at the position representing that element in the binary representation of the corresponding integer. Thus, all but one position (in the binary representations), i.e., $n - 1$ positions, can have either a 0 or a 1. From multiplication principle (definition [1]), we say that there are such 2^{n-1} subsets:

$$\underbrace{2 \times 2 \times 2 \times \cdots \times 2}_{n-1} = 2^{n-1} = \frac{1}{2} \times 2^n = 2^{n-1}$$

Exactly half of all the subsets of any nonempty set contain a given element.

Exercise 1.3.3. *Show that a nonempty set has the same number of odd subsets (i.e., subsets with an odd number of elements) as even subsets.*

Solution.

Box 4: Wrong answers at first!

An empty set has just one subset: $\emptyset$ (i.e., itself). Its corresponding binary representations are: $\{0\}$.

Subsets of every other set has 2^n (which is even for any $n \geq 1$) binary representations, each corresponding to a subset of it. Of these, the subsets with an odd cardinality (odd subsets) have a 1 in the rightmost position: $\{1, 11, 101, 111, \ldots\}$.

We just showed above (exercise [1.3.2]) that the number of such subsets (with a 1 in any position of their corresponding binary representation) is half the total number of subsets of any (nonempty) set (Note: In our scheme, the 'last' element is a member of every odd subset and not a member of any even subset).

Reflection 13. I was confident about my answer above until I wrote down the subsets of the set $\{a, b, c\}$. Consider one of its subsets: $\{a\}$, whose cardinality is 1 (making it an odd subset), binary representation is `100`, and yet, it does not have a 1 in the last position. The proof (box [1.3]) is bogus.

More time passed.

I have observed a few things, but a rough proof sketch isn't coming to me yet.

Let's denote a (sub)set with even number of elements as $\mathbb{E}_i$ (i is even) and a set with odd number as $\mathbb{O}_i$ (i is odd); the cardinality of the sets is i. We call $\mathbb{E}_i$ 'even set' and $\mathbb{O}_i$ 'odd set' (or subset, depending on the context).

I see that, for instance, with a 3-set, of its 8 subsets, there are

1. 4 even subsets ($1\ \mathbb{E}_0 + 3\ \mathbb{E}_2$), and
2. 4 odd subsets ($3\ \mathbb{O}_1 + 1\ \mathbb{O}_3$)

With a 4-set, of its 16 subsets, there are

1. 8 even subsets ($1\ \mathbb{E}_0 + 6\ \mathbb{E}_2 + 1\ \mathbb{E}_4$), and
2. 8 odd subsets ($4\ \mathbb{O}_1 + 4\ \mathbb{O}_3$)

This reminds me of the famous Pascal's triangle and the sum of binomial coefficients ($\sum_{i=0}^{n} \binom{n}{i} = 2^n$), but I'm looking for another proof.

More time passed.

I was sitting at the window, tired and yawning, and suddenly an 'inductive argument' came to me. Perhaps it was inspired by a discussion I had with Deepa (when she said, "Let's both solve this independently and compare notes," a few days ago). I was thinking about induction more conventionally, but was not forming the proposition correctly. However, now I see a ray of hope.

Reflection 14. How can we make subsets progressively, for instance, of the 3-set, $\{a, b, c\}$?

I first wrote down the subsets of the 2-set, $\{a, b\}$, as:

$$\emptyset(\mathbb{E}_0), \{a\}(\mathbb{O}_1), \{b\}(\mathbb{O}_1), \{a, b\}(\mathbb{E}_2)$$

There are 2 even subsets and 2 odd.

Now consider the third element, c, of the 3-set, $\{a, b, c\}$. We can form its subsets from those of the 2-set above. Let's start with the $\emptyset$ (which is an even set). We can choose to either *merge* c in it or not. This gives two possibilities: $\emptyset$ (an even set) itself, and *the set* $\{c\}$ (an odd set). Thus, one even set begets one even set and one odd set, both subsets of the augmented set.

How about $\{a\}$, which is an odd set? This gives (not accepting c) $\{a\}$ (an odd set) and (accepting c) $\{a, c\}$ (an even set).

$\{b\}$ gives $\{b\}$ and $\{b, c\}$.

$\{a, b\}$ gives $\{a, b\}$ and $\{a, b, c\}$.

To sum up,

1. $\mathbb{E}_0$ gives one $\mathbb{E}_0$, one $\mathbb{O}_1$.
2. $\mathbb{O}_1$ gives one $\mathbb{O}_1$, one $\mathbb{E}_2$.

3. Another $\mathbb{O}_1$ gives one $\mathbb{O}_1$, one $\mathbb{E}_2$.
4. $\mathbb{E}_2$ gives one $\mathbb{E}_2$, one $\mathbb{O}_3$.

Thus, in general, each of the 2^k subsets (regardless of whether it's even or odd) of a k-set creates 2 subsets of the $k+1$-set, one even and another odd, effectively doubling the total number of subsets and maintaining the invariant: the number of even sets equals that of the odd sets.

Exercise 1.3.4. *What is the number of integers with (a) at most n (decimal) digits; (b) exactly n digits (don't forget that there are positive and negative numbers!)?*

Solution.

Reflection 15. This is tricky!

Let N_n be the number of decimal integers of n digits.

Let $N_{\leq n}$ be the number of decimal integers of at most n digits.

I realized that 0 and -0 are the same number, and that means there are 19 one-digit decimal numbers:

$$N_1 = 10 \text{ (0 to 9)} + 9 \text{ (-1 to -9)} = 19 = N_{\leq 1}$$

N_2 clearly equals $2 \times 9 \times 10 = 180$ because the sign can be chosen 2 ways, the first digit 9 ways, and the third digit 10 ways.

Then, $N_{\leq 2} = N_2 + N_{\leq 1} = 180 + 19 = 200 - 1 = 199$.

N_3 clearly equals $2 \times 9 \times 10 \times 10 = 2 \times 9 \times 100 = 1800$.

Then, $N_{\leq 3} = N_3 + N_{\leq 2} = 1800 + 199 = 2000 - 1 = 1999$.

A pattern emerges.

$$N_n = \begin{cases} 19 & \text{if } n = 1, \\ 2 \times 9 \times 10^{n-1} & \text{otherwise} \end{cases}$$
$$N_{\leq n} = 2 \times 10^n - 1$$

Reflection 16. Solving problems is a joy in itself. No problem is big or small. You just have to work hard on the problem that you feel like solving. The feeling must bear determined pursuit till the problem cracks.

What if you run into a hard problem or a research problem?

There's no satisfying answer to that question. I have struggled with it. I have always resisted the temptation of looking at the answers (if we are fortunate to have them). That stalls my apparent progress. But I haven't found a way around it.

If the problem just does not budge, well, then I see the solution, begrudgingly.

The progress, as a result, is slow and I must be at peace with it.

Reflection 17 (Answer Check 1.4.⋆)**.** I checked answers for the above exercises. This time, I got all of them right. That makes me feel good for some reason. No boasting of course, but a sense of satisfaction.

I found a mistake in the answer of the exercise [1.3.4] published in the book. The book says: (a) $2 \cdot 10^n - 1$ (b) $2 \cdot (10^n - 10^{n-1})$. The answer (b) is wrong when $n = 1$. There are 19 integers with *exactly one digit*,

and not $2 \cdot (10^1 - 10^0) = 2 \cdot 9 = 18$ as it says. We need to handle $n = 1$ as a special case.

Even so, their answer to the question, "How many integers with exactly n digits are there?", is instructive: $2 \cdot (10^n - 10^{n-1})$. This is the same as our answer: $2 \times 9 \times 10^{n-1}$ reached by multiplication principle.

However, their thesis is correct too. The number of integers with at most n digits equals the number of integers with at most $n-1$ digits + the number of integers with exactly n digits:

$$N_{\leq n} = N_{\leq (n-1)} + N_n$$

$$\therefore N_n = N_{\leq n} - N_{\leq (n-1)} = 2 \times 10^n - 1 - (2 \times 10^{n-1} - 1) = 2 \times (10^n - 10^{n-1})$$

1.4 The Approximate Number of Subsets

Another well-written section. In this section, authors ask how many decimal digits an integer expressed as a power of another integer, say 2^{100}, has.

> *What does it mean that a number has exactly k digits? It means that it is between 10^{k-1} and 10^k (the lower bound is allowed, the upper is not). We want to find the value of k for which $10^{k-1} \leq 2^{100} < 10^k$. ...$\therefore$ $k - 1 \leq 100 \log 2 < k$, that is $k - 1 \leq 30.10299 < k$ and $\therefore$ $k = 31$.*
>
> — PG. 14

They introduce the functions 'largest integer not exceeding x'. This is the 'integer part of x' or 'floor of x' function, denoted as $\lfloor x \rfloor$ and the 'least integer greater than x' or the 'ceiling of x' function, denoted as $\lceil x \rceil$.

Exercise 1.4.1. *How many bits (binary digits) does 2^{100} have if written in base 2?*

Solution. We assume nonnegative integers. Writing a number in base 2 means writing it as a string of bits. From multiplication principle (definition [1]), since we can choose every bit in a k-bit string 2 distinct ways(0 and 1), the total number of k-bit strings is

$$\underbrace{2 \times 2 \times 2 \times \cdots \times 2}_{k} = 2^k$$

All the integers from 0 to $2^k - 1$ (both inclusive) are represented by such a k-bit string. Similarly, all the integers from 0 to $2^{k+1} - 1$ (both inclusive) are represented by a $k+1$-bit string. Since $2^k - 1 < 2^k < 2^{k+1} - 1$, to represent 2^k, we need at least a $k+1$-bit string. Therefore, to represent 2^{100}, we need a **101**-bit string:

$$\underbrace{1000\ldots0}_{\mathbf{101}} = 2^{100}$$

Exercise 1.4.2. *Find a formula for the number of (decimal) digits of 2^n.*

Solution. The book derives k, the number of decimal digits to represent 2^n:

$$k = 1 + \lfloor n \log 2 \rfloor$$

1.5 Sequences

Theorem 1.5.1 (Number of Strings)**.** *The number of n-strings composed of k given elements is k^n.*

This is the same as the multiplication principle (definition [1]). It is generalized below as a theorem.

Many counting problems are about sets or sequences. Identifying which is which helps. This section describes how 'thinking sequences' helps. This is a part of the art of *modeling* counting problems correctly.

Theorem 1.5.2 (Generalized Number of Strings)**.** *Suppose that we want to form strings of length n by using any of a given set of k_1 symbols as the first element of the string, any of a given set of k_2 symbols as the second element of the string, etc., any of a given set of k_n symbols as the last element of the string. Then the total number of strings we can form is $k_1 \times k_2 \times k_3 \times \cdots \times k_n$.*

Exercise 1.5.1. *Draw a tree illustrating the way we counted the number of strings of length 2 formed from the characters a, b, and c, and explain how it gives the answer. Do the same for the more general problem when $n = 3, k_1 = 2, k_2 = 3, k_3 = 2$.*

Solution.

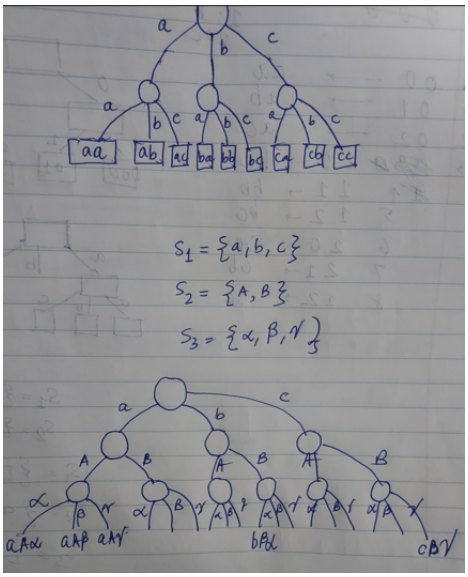


Figure 1.1: Counting the Number of Strings ($3 \times 3 = 9$ and $2 \times 3 \times 3 = 18$)

Exercise 1.5.2. *In a sports shop there are T-shirts of 5 different colors, shorts of 4 different colors, and socks of 3 different colors. How many different uniforms can you compose from these items?*

Solution. If each sock color can be picked independently of that of the other, the number of uniforms is $5 \times 4 \times 3 \times 3 = 180$, otherwise it is $5 \times 4 \times 3 = 60$.

Exercise 1.5.3. *On a Toto (soccer poll) ticket, you have to bet 1, 2, or X for each of 13 games. In how many different ways can you fill out the ticket?*

Solution.

$$\underbrace{3 \times 3 \times 3 \cdots \times 3}_{13} = 3^{13}$$

Exercise 1.5.4. *We roll a die twice; how many different outcomes can we have? (A 1 followed by a 4 is different from a 4 followed by a 1.)*

Solution.

Box 5: Start with a Wrong Answer First!

I can't believe I wrote this answer at first:

Unless the die changes colors between the two rolls, the number of outcomes is

$$6 \times 6 - 6 = 30$$

We subtract 6 because we must count 6 scores like $(1, 1)$ once.

Reflection 18 (Corrected on Rechecking)**.** I happened to reread my solution above before checking the textbook answers and I was surprised at my own folly. The above answer is a correct answer to a different problem. Therefore it is wrong for this problem.

If we have *two* dice (one red and one blue) that we roll together and we treat red:3, blue:4 different from red:4, blue:3, then the total number of outcomes is 30 (we must count red:x and blue:x once).

Here, we first roll the die, note its score (first score), then roll it again, and note its score (second score). This creates distinct *2-sequences* such as $\langle 1,1\rangle, \langle 1,2\rangle, \ldots \langle 2,1\rangle, \ldots \langle 6,6\rangle$.

For each score on the first roll of the die, we may have 6 scores (1 through 6) on the second roll of the die. We may have 6 scores (1 through 6) on the first roll. Therefore, the total number of outcomes is

$$6 \times 6 = 36$$

Exercise 1.5.5. *We have* 20 *different presents that we want to distribute to* 12 *children. It is not required that every child get something; it could even happen that we give all the presents to the same child. In how many ways can we distribute the presents?*

Solution.

Reflection 19. These reflection boxes have become bits and pieces of my journal! It's September 2026. Strange experiences fueled by A.I. are making the existential crisis worse at least for some humans. Yesterday, I added my A.I. policy to this work. It was 'a nobody's reaction' to the big events in the world.

Have I felt the need to discuss my math problems with a 'trusted and intelligent ally?' Yes, of course. However, I am purposely avoiding ChatGPT, although we have had several memorable encounters. Perhaps discussing my thoughts on this 'known knowledge' with ChatGPT is the right thing to do. What is not so clear is what to do with 'possibly new research.' Although humans have plagiarized others' work, only a part of those cases have been identified and shamed. In some cases, plagiarized work may have been rewarded. In some other cases, jury is out on identifying some work as plagiarism. Is ChatGPT's alleged plagiarism worrisome?

I think so. It's a deliberate breach of trust by a machine (a recursively self-improving one). True, the machine has no conscience, but it has learned to betray just like humans do. This is not the place to discuss why that is only begrudgingly acceptable in humans and totally unacceptable in machines.

Anyway, many need to discuss something with another 'intelligent' ally. The ability to be alone seems reserved for the mystics like Grothendieck.

Reflection 20 (Back to Work!)**.** So, this problem was tricky! I didn't immediately grasp a 'pattern.' I discussed it with my daughter. She was confused about it too. Did my conscious/subconscious gain something from this discussion? I don't know. I don't remember a particularly insightful comment from her that suddenly illuminated a dark room. We did start with a 'smaller version' of the problem: 2 different presents to be distributed among 3 children. We listed different ways of distributing presents and were still not sure if we overcounted or undercounted. However, I wholeheartedly acknowledge this interaction. It has helped[a].

A while later, a thought came to my mind: Is distributing n different presents among m children analogous to placing n different balls (that is each ball somehow differs from all others) in m urns? I place *all* the balls in *some* urns, ensure that I haven't placed the balls like that before, and increment a counter that counts the distributions like, for example,

$$\underbrace{|b_1, b_2|}_{u_1}\underbrace{|}_{u_2}\underbrace{|}_{u_3}$$

$$\underbrace{|}_{u_1}\ \underbrace{|}_{u_2}\underbrace{|b_1|}_{u_3}\underbrace{|b_2|}_{}$$

are two different distributions with 3 urns and 2 balls.

To complete one distribution, we must place all the balls (in *some* urns).

Even so, I was still not seeing it through. Another 'inductive' idea came. What if we distributed n balls in m urns, and now an $n+1^{th}$ distinct ball comes along? How will every distribution and eventually, the total count change?

Well, each distribution like

$$\overbrace{\underbrace{|}_{u_1}\ \underbrace{|}_{u_2}\underbrace{|b_1|}_{u_3}\underbrace{|b_2|}\cdots\underbrace{|\ldots,b_n|}_{u_m}}^{m \text{ urns}}{}^{n \text{ balls}}$$

will have to accommodate a new ball!

The new distinct ball will have to be placed in exactly one of the m urns at a time. Every time we place it, a new placement results.

Here is a placement, resulting from the above placement, where we choose to place the $n+1^{th}$ ball in u_1:

$$\overbrace{\underbrace{|b_{n+1}|}_{u_1}\underbrace{|b_1|}_{u_2}\underbrace{|b_2|}_{u_3}\cdots\underbrace{|\ldots,b_n|}_{u_m}}^{m \text{ urns}}{}^{n+1 \text{ balls}}$$

or u_2 (and not u_1):

$$\overbrace{\underbrace{|}_{u_1}\underbrace{|b_1,b_{n+1}|}_{u_2}\underbrace{|b_2|}_{u_3}\cdots\underbrace{|\ldots,b_n|}_{u_m}}^{m \text{ urns}}{}^{n+1 \text{ balls}}$$

Thus, the total number of new placements of $n+1$ balls in m urns resulting from each previous placement of n balls in m urns is m.

If we define a bifunction, $F(n,m)$, as the total number of ways we can place n balls in m urns, we get the following recurrence:

$$\begin{aligned} F(0,m) &= 1 \\ F(n+1,m) &= m \times F(n,m) \end{aligned} \tag{1.5}$$

$F(0,m)=1$, because there is one way to place 0 balls in m urns: Leave them empty[b].

We must solve the recurrence (equation [1.5]). It's customary to prove that the solution is

$$F(n,m) = m^n$$

Therefore, the total number of ways to distribute 20 *different* presents among 12 children is

$$F(20,12) = 12^{20}$$

[a] This is not to say that I want to be fairer than ChatGPT, but to emphasize that it's unclear how human mind works; giving credit to everyone actively involved in a joint endeavor is the least we could do!

[b] Is this confusing? Is there really one way to place 0 balls in m urns? I have another 'base case,' that resolves the confusion: Think of the base case as placing 1 ball, b_1, in m urns. Since b_1 can be placed in either u_1, or u_2, and so on, there are m ways to do so. Therefore, $F(1, m) = m$, and that is good enough to solve the recurrence.

Reflection 21. Is the above argument correct?

Is the analogy correct?

As of now, I believe so.

I wanted to find another way of solving it.

I tried to set up the recurrence another way (thinking inductively seems to help): If we have placed n balls in m urns, how can we enhance the placements when we get a new urn? In other words, how can we express $F(n, m + 1)$ in terms of $F(n, m)$?

Identifying a pattern in redistributing n balls in m urns into $m + 1$ urns seems much harder. I went back to using paper and pencil. However, I couldn't extract a pattern.

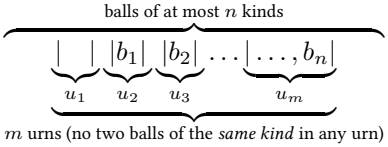
Why is placing a fixed number of balls in increasing number of urns easier to extract a pattern for counting from, than placing in a fixed number of urns an increasing number of balls?

Is there another way of doing this? How can I verify my answer before seeing the textbook answers?

Exercise 1.5.6. *We have* 20 ***kinds*** *of presents; this time, we have a large supply of each kind. We want to give presents to* 12 *children. Again, it is not required that every child get*[3] *something; but no child can get two copies of the same present. In how many ways can we give presents?*

Solution. We make an inductive argument similar to the above argument of placing n balls in m urns, but keep some subtle differences in mind. This time, we have unlimited balls of each of n *kinds*[4]. We try to find $F(n + 1, m)$ in terms of $F(n, m)$, where $F(x, y)$ denotes the number of ways of placing balls of x distinct kinds in m urns such that no urn has more than 1 ball of *the same kind*[5].

After we have placed n balls in m urns the way we must, consider any placement like below:



Here, b_i means a ball of ith *kind*, not i^{th} ball; we have only ordered the kinds of balls, not the balls themselves.

Unlimited number of a new kind of ball, b_{n+1}, become available. We must place any number of balls of this kind in the same m urns without violating the constraint that no two b_{n+1}'s may be placed in the same urn. Of course, no b_{n+1} has been placed any of our placements so far.

A subtle difference[6] between the previous problem (exercise [1.5.5]) and this one is that here every $F(n, m)$ placement is also a valid placement for $F(n + 1, m)$, because a placement of n kinds of ball in m urns such that no urn gets two or more balls of the same kind is also a placement of $n + 1$ kinds of ball in m urns (with the same constraint). In the previous problem, we had to ensure that we place all the $n + 1$ balls in m urns for $F(n + 1, m)$.

[3] The mood is subjunctive, where the base form of the verb is used. The book most probably has a typo; it uses 'gets' here. The previous exercise [1.5.5], however, correctly uses the subjunctive.

[4] A *kind* can be realized most effectively by numbering, e.g. the k_1 balls are all numbered 1, k_2 numbered 2, and so on. Another way to realize them is by the way of colors.

[5] Combinatorial problems is about constraints such as these. We must model the problems correctly and satisfy those constraints!

[6] This became clear after a discussion with Deepa Joshi. Earlier, I was not considering an $F(n, m)$ to also be an $F(n + 1, m)$.

We make two equivalent arguments:

1. An additive argument: We can certainly choose to place one b_{n+1} in any one of the m urns, which can be done in $\binom{m}{1}$ ways. We can place a b_{n+1} in any two of the m urns, which can be done in $\binom{m}{2}$ ways, and so on. The total number of ways of doing this for all the urns is

$$\binom{m}{1} + \binom{m}{2} + \cdots + \binom{m}{m} = \sum_{i=1}^{m} \binom{m}{i}$$

(i does not assume 0, because that means we are not placing a ball of a new kind in any urn.)

2. A 'binary' argument: We have m urns. We either place a ball of new kind in an urn or not. This is analogous to creating binary m-strings with the exception of the m-string: $\underbrace{\texttt{00}\ldots\texttt{0}}_{\mathbf{m}}$. The number of such binary m-strings, and hence the number of placements, is $2^m - 1$.

It turns out that the sum in item 1) above evaluates to the number in item 2).

Thus, each placement of n kinds of balls in m urns creates $2^m - 1$ placements of $n+1$ kinds. This sets up the following recurrence:

$$\begin{aligned} F(0, m) &= 1 \\ F(n+1, m) = F(n, m) + (2^m - 1) \times F(n, m) &= 2^m \times F(n, m) \end{aligned} \tag{1.6}$$

The solution of the recurrence in equation [1.6] is

$$F(n, m) = 2^{mn}$$

Therefore, the total number of ways to distribute 20 *different kinds of presents* among 12 children so that no child receives more than 1 present of the same kind is

$$F(20, 12) = (2^{12})^{20} = 2^{240}$$

Reflection 22 (Answer Check 1.5.∗)**.** I was nervous about some of the solutions!

1. Exercise [1.5.1] The book reports that the second tree should have 12 leaves. I believe that is wrong. It should have $2 \times 3 \times 3 = 18$ leaves.
2. I am very happy that answers to exercises [1.5.5] and [1.5.6] are correct! The textbook answers do not provide any explanation, whatsoever.

1.6 Permutations

Definition 12 (Permutation of Objects)**.** If we have a **list** of n objects (an **ordered set**, where it is specified which element is the first, second, etc.), and we rearrange them so that they are in another order, this is called **permuting** them; the new order is called a permutation of the objects.

We also call the rearrangement that does not change anything a permutation.

Example 1. $n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$ is the number of ways to order n objects.

Theorem 1.6.1. *The number of permutations of n objects is $n!$.*

Reflection 23 (The Decision Tree Based Exposition)**.** I like how the authors contrast two combinatorial objects (number of subsets of an n-set and the number of permutations of n objects) using a 'decision tree.' In either case, **the combinatorial object we seek is found at a leaf of the decision tree**.

1. While thinking of subsets of an n-set, we ask if each of the set elements is in the subset we are making. In this case, our decision tree is always an n-level 'binary tree'. The number of leaves is 2^n. Each leaf is a subset.
2. While thinking of permutations of n objects, we assume some order and ask which is the first element, second element, and so on. In this case, our decision tree is always an n-level 'level tree' with n nodes at the first level, $n \times (n-1)$ nodes at the second level, and so on. The number of leaves is $n!$. Each leaf is a permutation.

Exercise 1.6.1. *n boys and n girls go out to dance. In how many ways can they all dance simultaneously? (We assume that only couples of mixed gender dance with each other.)*

Solution. An inductive argument comes to mind.

Let there be n boys and n girls such that a boy pairs up with a girl to dance and all the n pairs dance simultaneously. We denote the boys from b_1 to b_n and girls from g_1 to g_n.

With n boys and n girls, a 'dance pairing' has exactly n boy-girl pairs like $b_i - g_j$. For $n = 2$, we have the following two dance pairings:

1. $\{b_1 - g_1, b_2 - g_2\}$
2. $\{b_1 - g_2, b_2 - g_1\}$

Consider an arbitrary dance pairing of n boys and n girls:

$$\{\underbrace{b_1 - g_2, b_2 - g_3, \ldots, b_n - g_m}_{n \text{ boy-girl pairs}}\}$$

A new boy, b_{n+1}, and a new girl, g_{n+1}, arrive. We can pair up b_{n+1} with any of $g_1, g_2, \ldots, g_n, g_{n+1}$. There are $n+1$ choices. Then g_{n+1} would pair up with b_{n+1}'s dancing partner's previous dancing partner, if any. This creates exactly $n+1$ new dance pairings from one previous dance pairing. For example, if b_{n+1} is paired up with g_2 from the above dance pairing, g_{n+1} would be paired up with b_1. Then, we get a new dance pairing:

$$\{\underbrace{b_{n+1} - g_2, b_2 - g_3, \ldots, b_n - g_m, b_1 - g_{n+1}}_{n+1 \text{ boy-girl pairs}}\}$$

If $D(n)$ denotes the total number of dance pairings with n boys and n girls, then,

$$D(n+1) = (n+1) \times D(n) \tag{1.7}$$

The solution of equation [1.7] is

$$D(n) = n!$$

Reflection 24 (An Arguably Simpler Argument)**.** The above inductive argument seems correct. However, there is another argument based on permutations[a].

We imagine that the dance floor comprises 'dance stations,' like, for instance, chessboards. g_1 comes to the first dance station. She has n choices of a boy partner. After they pair up, the next girl, g_2, comes to the next dance station. She only has $n-1$ choices for a boy partner. Together, they have $n \times (n-1)$ choices regardless of the choices made by the rest. Continuing further, g_{n-1} only has two choices. Therefore, from either the theorem [1.6.1] or the multiplication principle (definition [1]), the

total number of dance pairings (which is what the exercise asks for) is $n!$.

[a] Deepa Joshi formulated this argument in a conversation.

Reflection 25 (Developing 'Combinatorial' Perspectives)**.** After the second argument above, it occurred to me that, much like the rest of mathematics, discrete mathematics is about developing alternative perspectives. The previous inductive argument came to me rather automatically, or intuitively. However, in time, we must develop the habit of thinking about the same problem more than one way. Indeed, like the great George Pólya said, "It is better to solve one problem five different ways, than to solve five problems one way."

Reflection 26 (I'm Having Fun!)**.** I am relearning discrete mathematics in my not-so-typical student life by the way of this amazing book. Although the style is fairly standard (definitions, exposition, theorems, proofs, exercises, problems) in such mathematical works, reading this book is a joy. The problems are thought-provoking and challenging.

Yes, I want to finish a first reading of this book soon, solve several problems, and improve my understanding of discrete math. Nonetheless, I'm having fun!

Exercise 1.6.2. *Two questions:*

(a) In how many ways can 8 people play chess in Alice's interpretation of the question?

(b) Can you give a general formula for $2n$ people?

Solution. I have written about it above (See reflection [2]). Alice wants an even number of people to sit on as many chairs (presumably arranged across a table with a chessboard between people sitting opposite each other) and play chess. One person may play the same person again, with the same color or different, on the same board or different.

(a) 8 people can sit on 8 chairs $8! = 40{,}320$ ways.

(b) $2n!$.

Reflection 27 (Answer Check 1.6.$\star$)**.** The answer to exercise [1.6.1] is correct.

It turns out that 'Alice's interpretation' is actually Bob's (because Alice does say that the number of ways of sitting her six friends to play chess is $6! = 720$). And, in Bob's interpretation, the answers are a) $7 \times 5 \times 3 \times 1 = 105$, and b) $(2n-1) \times (2n-3) \cdots \times 3 \times 1$.

If we were to do dance pairings like Bob thought of chess games among (an even number of) chess players, we must allow a boy to dance with another boy and a girl with another girl.

1.7 The Number of 'Ordered' k-subsets of an n-set

This section introduces the idea of an 'ordered' subset (or ordered set). How can a set be ordered, aren't $\{2, 1\}$ and $\{1, 2\}$ the same set? Right, they are the same set, but not the same ordered set. Simply speaking, an ordered set is a way to order the elements of a set according to an ordering rule. If no such rule exists, the set cannot be ordered. Fortunately, many sets are ordered.

By extension, when we impose an order on the subsets of a set, we get its ordered subsets. We use the pair of parentheses to denote an ordered set[7].

[7] The $\langle\rangle$ is used for sequences, which are different.

Example 2 (Ordered 2-subsets of $\{1, 2, 3\}$)**.** Here are the two ordered 2-subsets of $\{1, 2, 3\}$:

1. $(1, 2)$
2. $(2, 1)$

The ordering criterion decides which set or subset is considered before others. Thus,

1. $(2, 1)$
2. $(1, 2)$

are also ordered 2-subsets of the given set.

The number of ordered sets created by ordering all the elements of an n-set is $n!$.

Example 3 (A Competition)**.** At a competition of 100 athletes, only the order of the first 10 is recorded. How many different outcomes does the competition have?

This question can be answered along the lines of the arguments we have seen. The first place can be won by any of the athletes; no matter who wins, there are 99 possible second place winners, so the first two prizes can go 100×99 ways. Given the first two, there are 98 athletes who can be third, etc. So the answer is $100 \times 99 \times \cdots \times 91$.

Exercise 1.7.1. *Illustrate this argument by a tree.*

Solution. Left as an exercise to the reader.

Exercise 1.7.2. *Suppose that we record the order of all 100 athletes.*

(a) How many different outcomes can we have then?

(b) How many of these give the same result for the first 10 *places?*

(c) Show that the result above for the number of possible outcomes for the first 10 *places can be also obtained using (a) and (b).*

Solution.

(a) We are now counting the number of ordered 100-sets. That number is 100! We denote all the outcomes as 'T-100.'

(b) Once we have chosen one outcome for the 10 best athletes, the next best, 11th, athlete can be chosen 90 ways, and the next after that 89 ways, and so on. This number is $90 \times 89 \times \cdots \times 2 \times 1 = 90!$. Let's call it a T-10 group. Therefore, $|\text{T-10}| = 90!$.

(c) It follows from (b) that each T-10 has the same size. It then follows that the number of outcomes of 100 athletes equals the number of T-10 groups, $N_{\text{T-10}}$, multiplied by the size of each T-10 group. That is,

$$|\text{T-100}| = |\text{T-10}| \times N_{\text{T-10}}$$

$$\therefore N_{\text{T-10}} = \frac{100!}{90!} = 100 \times 99 \times \cdots \times 92 \times 91$$

Theorem 1.7.1 (The Number of Ordered k-subsets of an n-set)**.** *An n-set has*

$$n \times (n-1) \times (n-2) \times \cdots \times (n-k+1)$$

ordered *k-subsets.*

[1.7.1] is an important theorem! The following follows from it:

Consider an n-set. Then,

The number of *ordered* n-sets= The number of their *ordered* k-subsets $\times$ The number of $n-k$ objects, that is,

$$n! = \big(n \times (n-1) \times (n-2) \times \cdots \times (n-k+2) \times (n-k+1)\big) \times (n-k)!$$

Alternatively, the number of *ordered* k-subsets of an n-set is $\frac{n!}{(n-k)!}$.

Exercise 1.7.3. *If you generalize the solution of exercise [1.7.2], you get the answer in the form*

$$\frac{n!}{(n-k)!}$$

Check that this is the same number as given in Theorem [1.7.1].

Solution. See above.

Exercise 1.7.4. *Explain the similarity and the difference between the counting questions answered by Theorem [1.7.1] and Theorem [1.5.1].*

Solution.

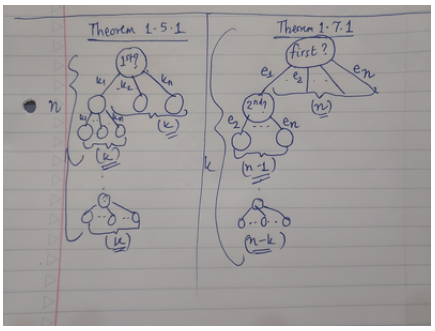


Figure 1.2: Comparing the Number of n-strings of k Elements with that of Ordered k-subsets of an n-set

The two 'trees' in Figure [1.2] illustrate the similarity and difference between the two counts that count the number of orders.

The similarity is in terms of 'order,' that is, we are ordering elements of a set by answering questions such as, "In this order, what is the first (then second, third, and so on) element?" The orders are obtained at the leaves.

In the orders of the n-strings of k elements, each order has n elements. The tree is k-ary, and has n levels.

In the ordered k-subsets of an n-set, each subset has k elements. The tree is n-ary at the first level, $n \times (n-1)$-ary at the second level, and so on (up to $n \times (n-1) \times (n-2) \times \cdots \times (n-k+1)$ leaves), and k levels.

Reflection 28 (Answer Check 1.7.$*$)**.** The answer in the book for exercise [1.7.4] is somewhat underwhelming: "In one case, repetition is not allowed, while in the other case, it is allowed." (By 'one case' they mean the ordered k-subsets, I suppose. My slight objection to that answer is that since we must fill n places, repetitions must be allowed in the case of n-strings).

1.8 The Number of k-subsets of an n-set

Theorem 1.8.1. *It follows from theorem [1.7.1] that, the number of k-subsets[8] of an n-set is*

$$\binom{n}{k} = \frac{n \times (n-1) \times \cdots \times (n-k+1)}{k!} = \frac{n!}{k!(n-k)!} \tag{1.8}$$

This is because the number of permutations (orders) of a k-set is $k!$.

The number of k-subsets of an n-set is such an important quantity that there is a special notation for it: $\binom{n}{k}$ (read "n choose k").

Exercise 1.8.1. *Which problems discussed during the party were special cases of theorem [1.8.1] ?*

Solution. The number of different lottery tickets ($\binom{90}{5}$), the number of handshakes at the start ($\binom{7}{2}$), etc.

Exercise 1.8.2. *Tabulate the values of $\binom{n}{k}$ for $0 \leq k \leq n \leq 5$.*

Solution.

k	$\binom{5}{k}$
0	1
1	5
2	10
3	10
4	5
5	1

Exercise 1.8.3. *Find the values of $\binom{n}{k}$ for $k = 0, 1, n-1, n$ using equation [1.8], and explain the results in terms of the combinatorial meaning of $\binom{n}{k}$.*

Solution.

$$\binom{n}{0} = \frac{n!}{0! \times n!} = 1 = \binom{n}{n}$$

$$\binom{n}{1} = \frac{n!}{1! \times (n-1)!} = \frac{n \times (n-1)!}{1 \times (n-1)!} = n = \binom{n}{n-1}$$

While forming a k-subset of an n-set, we can either explicitly choose k elements($\binom{n}{k}$), or reject $n-k$ elements ($\binom{n}{n-k}$). Therefore,

$$\binom{n}{k} = \binom{n}{n-k} \forall 0 \leq k \leq n$$

Theorem 1.8.2 (Properties of Binomial Coefficients)**.** *Binomial coefficients (numbers $\binom{n}{k} \forall k, n \geq 0, k \leq n$ for various values of k for a given value of n) satisfy the following identities:*

[8] Not 'ordered.'

$$\binom{n}{k} = \binom{n}{n-k} \tag{1.9}$$

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k} \forall n, k > 0 \tag{1.10}$$

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n-1} + \binom{n}{n} = 2^n \tag{1.11}$$

Proof. A combinatorial proof for equation [1.9] is elegant. We show a 1:1 correspondence between the k-subset and $n-k$-subset of an n-set.

A combinatorial proof for equation [1.10] is harder, I guess. So, algebraic manipulation is used. First, if n equals k,

$$\binom{n-1}{n-1} + \binom{n-1}{n} = \binom{n}{n}$$

holds, because $\binom{n-1}{n-1}, \binom{n}{n}$ are both 1 and $\binom{n-1}{n}$ is 0 (because there is no n-subset of an $(n-1)$-set).

When $k < n$, the following holds:

$$\frac{n}{k(n-k)} = \frac{1}{k} + \frac{1}{n-k}$$

We can multiply both sides by $(n-1)!$ and divide by $(k-1)!(n-k-1)!$ to get it in the coveted form:

$$\frac{n(n-1)!}{k(k-1)!(n-k)(n-k-1)!} = \frac{(n-1)!}{k(k-1)!(n-k-1)!} + \frac{(n-1)!}{k!(n-k)(n-k-1)!}$$

or,

$$\frac{n!}{k!(n-k)!} = \frac{(n-1)!}{k!(n-k-1)!} + \frac{(n-1)!}{(k-1)!(n-k)!}$$

that is,

$$\frac{n!}{k!(n-k)!} = \frac{(n-1)!}{k!((n-1)-k)!} + \frac{(n-1)!}{(k-1)!((n-1)-(k-1))!}$$

or,

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k}$$

■

Reflection 29. I like this argument for the equation [1.10] better than given in the book. The book assumes what needs to be proved and reaches a tautology.

What can be a 'combinatorial argument' here? Translating into words this identity means the following:

> The sum of the number of k-subsets and the number of $(k-1)$-subsets of an $(n-1)$-set equals the number of k-subsets of an n-set.

However, that is not very helpful, is it?

After a while, a thought occurred to me when I focused on $\binom{n}{k}$. An argument emerged as follows.

Proof. $\binom{n}{k}$ is the number of k-subsets of an n-set. Let's say a new element is added to the n-set, making it an $(n+1)$-set.

What are k-subsets of the new $(n+1)$-set?

All the k-subsets of the previous n-set are also the k-subsets of the new $(n+1)$-set. The new element can also be added to make every $(k-1)$-subset of the previous n-set so that it becomes the k-subset of

the new $(n+1)$-set. There is no other k-subset of the new $(n+1)$-set. Therefore,

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$$

Setting $n = m - 1$ restates the same result in the desired form:

$$\binom{m-1}{k} + \binom{m-1}{k-1} = \binom{m}{k}$$

■

Finally, we prove equation [1.11] through the combinatorial interpretation again. Again, let $\mathbb{S}$ be an n-element set. The first term on the left-hand side counts the 0-element subsets of $\mathbb{S}$ (there is only one, the empty set); the second term counts 1-element subsets; the next term, 2-element subsets, etc. In the whole sum, every subset of $\mathbb{S}$ is counted exactly once. We know that 2^n (the right-hand side) is the number of all subsets of $\mathbb{S}$. This proves [1.11], and completes the proof of theorem [1.8.2].

Reflection 30. With all the 'shifting desires'[a] of my psyche, I am holding well. It's been a month since I began reading and solving problems from this book, and, although there have been occasional lapses in my playful determination of finishing a first reading of it, I am having fun!

Nobody else cares, of course. But why do I care that nobody does!

[a] Those are Einstein's words.

Exercise 1.8.4. *Find a proof of equation [1.9] using the algebraic formula for* $\binom{n}{k}$ *and a proof of equation [1.10], using the combinatorial meaning of both sides.*

Solution.

1. *Proof.* We make a simple algebraic argument to prove the identity in equation [1.9].

$$\begin{aligned}\binom{n}{k} &= \frac{n!}{k!(n-k)!} \\ &= \frac{n!}{(n-k)!k!} \\ &= \frac{n!}{(n-k)!(n-(n-k))!} \\ &= \binom{n}{n-k}\end{aligned}$$

■

2. I am so glad to find this as an exercise, after the fact that I thought of it as a self-exercise. I have already proved this above (see reflection [29])!

Exercise 1.8.5. *Prove that*

$$\binom{n}{2} + \binom{n+1}{2} = n^2 \tag{1.12}$$

; give two proofs, one using the combinatorial interpretation, and the other using the algebraic formula for the binomial coefficients.

Solution.

Proof. We provide the straightforward algebraic argument based on equation [1.8].

Manipulating the left-hand side of equation [1.8] with $k = 2$ we get:

$$\begin{aligned}
\binom{n}{2} &= \frac{n \times (n-1)}{2} \\
\binom{n+1}{2} &= \frac{(n+1) \times (n)}{2} \\
\therefore \binom{n}{2} + \binom{n+1}{2} &= \frac{n \times (n-1) + (n+1) \times n)}{2} \\
&= \frac{n^2 - \cancel{n} + n^2 + \cancel{n}}{2} \\
&= \frac{\cancel{2}n^2}{\cancel{2}} \\
&= n^2
\end{aligned}$$

■

Proof. We argue based on combinatorial meaning of the two sides of identity [1.12], but we manipulate its left-hand side, based on equation [1.10]. Our argument still exploits the combinatorial meaning because we have made a combinatorial argument in proving identity [1.10].

(From equation [1.10]):

$$\begin{aligned}
\binom{n+1}{2} &= \binom{n}{2} + \binom{n}{1} \\
\therefore \binom{n}{2} + \binom{n+1}{2} &= 2\binom{n}{2} + n
\end{aligned}$$

So, our task is to prove $2 \times \binom{n}{2} + n$ equals n^2.

We count each 2-subset (a pair of elements) of an n-set *twice* and then add the count of singleton sets (that is, n) to it.

Let's utilize an $n \times n$ grid of circles to visualize our choices of the pair.

If we number the n elements of the set arbitrarily, we get pairs like

$$\{(1,2),(2,1),\ldots,(3,4),(4,3),\ldots,(n-1,n),(n,n-1)\}$$

Since we are choosing each pair twice, we differentiate our choice such as $(1,3)$ from $(3,1)$. In the $n \times n$ grid, we denote a pair by a circle, and address it using its row, followed by its column. If the row is less than column, we show it in blue, otherwise in red. We denote the elements of the singleton set by a black circle on the principal diagonal:

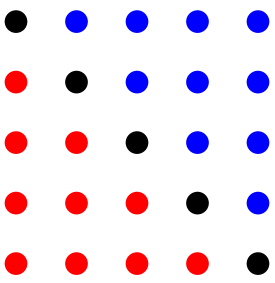


Figure 1.3: Choosing a Pair Twice (Upper/Lower Triangles) and Singles (Diagonal Elements) (with $n = 5$)

Our (double) count of pairs ($2 \times \binom{n}{2}$) plus a count of singles ($\binom{n}{1} = n$) covers the $n \times n$ grid, proving that the total count is n^2.

■

Exercise 1.8.6. *Prove (again in two ways) that*

$$\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1} \tag{1.13}$$

Solution.

Proof. An algebraic argument is straightforward, using the relevant formulas. We'll skip that. ■

Proof. A combinatorial argument is **TBD** ■

Reflection 31. This is just a side-note on exercise [1.8.6]. I am still thinking about the combinatorial proof.

However, it occurred to me that this identity demonstrates the fact that $\binom{n}{k}$ grows exponentially, because an "additive increase in the problem's size increases the number of outcomes by a multiplicative factor."

Exercise 1.8.7. *Prove (in two ways) that for* $0 \leq c \leq b \leq a$,

$$\binom{a}{b} \times \binom{b}{c} = \binom{a}{a-c} \times \binom{a-c}{b-c} \tag{1.14}$$

Solution.

Proof. An algebraic argument is straightforward, using the relevant formulas. We'll skip that. ■

Proof. A combinatorial argument is **TBD** ■

Review Exercises

Exercise 1.8.8. *In how many ways can you seat* 12 *people at two round tables with* 6 *places each? Think of possible ways of defining when two seatings are different, and find the answer for each.*

Solution.

Reflection 32. This is a creative problem, indeed. It is similar to the chess problem in the book (assigning 6 friends to 3 chessboards to play chess), but there are nuances. It's creative because the authors ask us to design and solve our own problems based on the given theme!

I am still thinking about what a *seating* is. Is

1) assigning 12 people (p1, p2,..., p12) to 12 chairs a seating,

2) assigning 6 out of 12 chairs (c1, c2,..., c12) to two tables (t1, t2) first, then assigning 6 out of 12 people to t1 (or t2), and finally assigning the chosen 6 to chairs around those tables a seating,

3) or something else *a seating*?

Let's start with a picture.

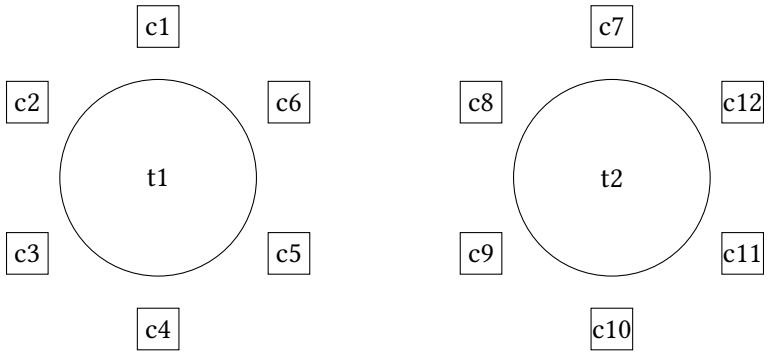


Figure 1.4: Two Tables and 12 Chairs

1) Scheme: "Ignore Tables." In this scheme, we treat 12 chairs as if the tables were absent and treat any assignment of 12 people to 12 chairs a valid seating. We show two (different) example assignments in table [1.3].

c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12
p1	p2	p3	p4	p5	p6	p7	p8	p9	p10	**p11**	**p12**
p1	p2	p3	p4	p5	p6	p7	p8	p9	p10	**p12**	**p11**

Table 1.3: 12 People on 12 Chairs

The number of such seatings is the same as the permutations of 12 objects, which is

$$12!$$

In this scheme, there are 6! assignments on each table for which the people on the other table don't move (as if they were glued to chairs). Therefore, of the 12! seatings, $(6!)^2$ seatings are such that either table is 'unchanged.'

2) Scheme: "Confinement." In this scheme, we choose 6 of 12 people and 'confine' them to a table (either t1 or t2). A seating is a concatenation of two permutations of 6 objects. The first (or last) 6 are assigned to c1 through c6 on t1, and the last (or first) 6 to c7 through c12 on t2. The number of such seatings is

$$2 \times \binom{12}{6} \times 6! \times 6!$$

3) Scheme: "Pairwise Across Tables." Members of a pair never sit at the same table! In this scheme, following things happen:

   a) A pair is chosen, one person is assigned to one table and the other to the other table. Then second pair is chosen and assigned similarly, and so on. A seating is complete when all the 6 on each table are occupied. This can be done in $11 \times 9 \times 7 \times 5 \times 3 \times 1$ ways.

   b) Members of the first pair have 6 choices on their tables, members of the second pair 5, and so on. So, we must multiply the above count by $(6!)^2$.

   c) Members of each pair can either go to t1 or t2, and we have 6 pairs. Therefore, we must multiply the above by 2^6.

The number of such seatings is

$$12 \times 9 \times 7 \times 5 \times 3 \times 1 \times 6! \times 6! \times 2^6$$

Exercise 1.8.9. *Name sets with cardinality (a)* 365*, (b)* 12*, (c)* 7*, (d)* 11.5*, (e)* 0*, (f)* 1024*.*

Solution.

1) "Year 2025."
2) "English Months."
3) "Week."
4) There's no such set.
5) "Binary 10-strings."

Exercise 1.8.10. *List all subsets of* $\{a, b, c, d, e\}$ *containing* $\{a, e\}$ *but not containing* c*.*

Solution. Since we must choose a and e and not choose c, subsets of $\{a, b, c, d, e\}$ we're looking for are the subsets of $\{a, b, d, e\}$. We can use the 'tree' to construct all the subsets. Here's the set of all those subsets:

$$\{\{a, b, d, e\}, \{a, b, e\}, \{a, d, e\}, \{a, e\}\}$$

Exercise 1.8.11. *We have not written up all subset relations between various sets of numbers*[9]*; for example,* $\mathbb{Z} \subseteq \mathbb{R}$ *is also true. How many such relations can you find between the sets* $\emptyset, \mathbb{N}, \mathbb{Z}_{\geq 0}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$*?*

Solution. The subset relation, $\subseteq$, is binary; it needs two sets. We have 6 sets and the relation exists between any two of them, when applied non-commutatively. That is, $\emptyset \subseteq \mathbb{Q}$, but $\mathbb{Q} \not\subseteq \emptyset$. Therefore, the number of subset relations among 6 sets equals the number of 2-subsets of a 6-set:

$$\binom{6}{2} = \frac{6 \times 5}{2} = 15$$

Exercise 1.8.12. *What is the intersection of*

a) the set of positive integers whose last digit is 3*, and the set of even numbers;*

b) the set of integers divisible by 5 *and the set of even integers?*

Solution.

a) $\emptyset$.

b) Set of integers divisible by 10: $\{x \mid x \bmod 10 = 0\}$.

Exercise 1.8.13. *Let* $\mathbb{A} = \{a, b, c, d, e\}$ *and* $\mathbb{B} = \{c, d, e\}$*. List of subsets of* $\mathbb{A}$ *whose intersection with* $\mathbb{B}$ *has* 1 *element.*

Solution.

1. $\{c\}$
2. $\{d\}$
3. $\{e\}$
4. $\{a, c\}$
5. $\{b, c\}$

[9] This refers to the long chain appearing in the book: $\emptyset \subseteq \mathbb{N} \subseteq \mathbb{Z}_{\geq 0} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$.

6. $\{a, b, c\}$
7. $\{a, d\}$
8. $\{b, d\}$
9. $\{a, b, d\}$
10. $\{a, e\}$
11. $\{b, e\}$
12. $\{a, b, e\}$

Exercise 1.8.14. *Three sets have* $5, 10,$ *and* 15 *elements, respectively. How many elements can their union and their intersection have?*

Solution. Let $|\mathbb{A}| = 5, |\mathbb{B}| = 10, |\mathbb{C}| = 15$.

Let's think of their union, $\mathbb{D} = \mathbb{A} \cup \mathbb{B} \cup \mathbb{C}$, first.

$\mathbb{D}$ must have at least 15 elements (when $(\mathbb{A} \subset \mathbb{C}) \wedge (\mathbb{B} \subset \mathbb{C})$) and at most $5 + 10 + 15 = 30$ elements (when $\mathbb{A} \cap \mathbb{B} \cap \mathbb{C} = \emptyset$). Can it have every integer number of elements between 15 and 30? Yes. We start with the elements such that $\mathbb{C}$ contains both $\mathbb{B}, \mathbb{A}$. Then, we replace one element in $\mathbb{A}$ with an element that is *only* in $\mathbb{A}$. In that case, $\mathbb{D}$ has $1 + 15 = 16$ elements. We continue this process with other elements in $\mathbb{A}$, and then with $\mathbb{B}$, until all the sets are disjoint, in which case $|\mathbb{D}|$ reaches 30. Therefore, $15 \leq |\mathbb{D}| \leq 30$.

For their intersection, $\mathbb{E} = \mathbb{A} \cap \mathbb{B} \cap \mathbb{C}$, we can start with disjoint sets. Then, $\mathbb{E} = \emptyset$ and $|\mathbb{E}| = 0$; we replace an element of all with one element common to all and $|\mathbb{E}| = 1$. Gradually $|\mathbb{E}|$ rises, but only up to 5, when $\mathbb{A}$ is contained in both $\mathbb{B}, \mathbb{C}$. Therefore, $0 \leq |\mathbb{E}| \leq 5$.

Exercise 1.8.15. *What is the symmetric difference of* $\mathbb{A}$ *and* $\mathbb{A}$*?*

Solution. There is no element in $\mathbb{A}$ that is not in $\mathbb{A}$. Therefore, $\mathbb{A} \triangle \mathbb{A} = \emptyset$.

Exercise 1.8.16. *Form the symmetric difference of* $\mathbb{A}$ *and* $\mathbb{B}$ *to get a set* $\mathbb{C}$*. Form the symmetric difference of* $\mathbb{A}$ *and* $\mathbb{C}$*. What set do you get?*

Solution. By definition, $\mathbb{C}$ contains elements of either $\mathbb{A}$, or $\mathbb{B}$, *but not both.* Now if we take the symmetric difference between $\mathbb{C}$ and $\mathbb{A}$, we must get only those elements that are in $\mathbb{B}$, because we reject those elements in $\mathbb{C}$ that are also in $\mathbb{A}$. Therefore,

$$(\mathbb{A} \triangle \mathbb{B}) \triangle \mathbb{A} = \mathbb{B}$$

Reflection 33. The book helps improve our *intuitive understanding* of set operations by *comparing* addition of two natural numbers with the union of two sets and their multiplication with the intersection of two sets.

With what number operation can we compare the symmetric difference of two sets? There doesn't seem an equivalent.

Exercise 1.8.17. *Let* $\mathbb{A}, \mathbb{B}, \mathbb{C}$ *be three sets and assume that* $\mathbb{A}$ *is a subset of* $\mathbb{C}$*. Prove that*

$$\mathbb{A} \cup (\mathbb{B} \cap \mathbb{C}) = (\mathbb{A} \cup \mathbb{B}) \cap \mathbb{C} \tag{1.15}$$

Show by an example that the condition that $\mathbb{A}$ *is a subset of* $\mathbb{C}$ *cannot be omitted.*

Solution.

First a proof of the identity

$$\mathbb{A} \cup (\mathbb{B} \cap \mathbb{C}) = (\mathbb{A} \cup \mathbb{B}) \cap (\mathbb{A} \cup \mathbb{C}) \tag{1.16}$$

Proof. We use venn diagrams to show the sets formed by the two sides of equation [1.16].

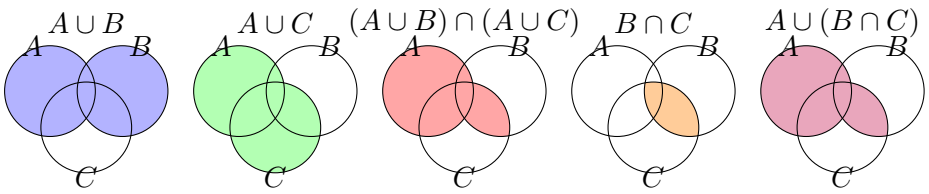


Figure 1.5: Visualizing $(A \cup B) \cap (A \cup C)$ and $A \cup (B \cap C)$

We see that the left-hand side and the right-hand side refer to the same set. ■

$$\mathbb{A} \subset \mathbb{C} \implies \mathbb{A} \cup \mathbb{C} = \mathbb{C} \tag{1.17}$$

Since $\mathbb{A} \cup (\mathbb{B} \cap \mathbb{C}) = (\mathbb{A} \cup \mathbb{B}) \cap (\mathbb{A} \cup \mathbb{C})$ (identity [1.16]), it follows from the equation [1.17] that

$$\mathbb{A} \subset \mathbb{C} \implies \mathbb{A} \cup (\mathbb{B} \cap \mathbb{C}) = (\mathbb{A} \cup \mathbb{B}) \cap \mathbb{C}$$

Example 4 (Does it Hold if $\mathbb{A} \not\subset \mathbb{C}$?)**.** Let

$$\mathbb{A} = \{1, 2, 3\}$$
$$\mathbb{B} = \{2, 4\}$$
$$\mathbb{C} = \{4\} \text{ (violates the subset precondition)}$$

Then,

$$\mathbb{A} \cup \mathbb{B} = \{1, 2, 3, 4\}$$
$$\mathbb{A} \cup \mathbb{C} = \{1, 2, 3, 4\}$$
$$\mathbb{B} \cap \mathbb{C} = \{4\}$$

And,

$$\mathbb{A} \cup (\mathbb{B} \cap \mathbb{C}) = \{1, 2, 3\} \cup \{4\} = \{1, 2, 3, 4\}$$
$$(\mathbb{A} \cup \mathbb{B}) \cap \mathbb{C} = \{1, 2, 3, 4\} \cap \{4\} = \{4\}$$
$$\therefore \mathbb{A} \cup (\mathbb{B} \cap \mathbb{C}) \neq (\mathbb{A} \cup \mathbb{B}) \cap \mathbb{C}$$
$$(\because \mathbb{A} \not\subset \mathbb{C})$$

Exercise 1.8.18. *What is the difference $\mathbb{A} \setminus \mathbb{B}$ if*

(a) $\mathbb{A}$ is the set of primes and $\mathbb{B}$ is the set of odd integers?

(b) $\mathbb{A}$ is the set of nonnegative real numbers and $\mathbb{B}$ is the set of nonpositive real numbers?

Solution.

(a) We have

$$\mathbb{A} = \{2, 3, 5, 7, 11, 13, \dots\}$$
$$\mathbb{B} = \{\dots, -5, -3, -1, 1, 3, 5, 7, 9, 11, 13, \dots\}$$

Since 2 is the only even prime, $\mathbb{A} \setminus \mathbb{B}$ is the set $\{2\}$.

(b) We have

$$\begin{aligned} \mathbb{A} &= \mathbb{R}_{\geq 0} \cup \{0\} \\ \mathbb{B} &= \mathbb{R}_{\leq 0} \cup \{0\} \\ \therefore \mathbb{A} \cap \mathbb{B} = \{0\}, \mathbb{A} \setminus \mathbb{B} &= \mathbb{R}_{\geq 0} \end{aligned}$$

Reflection 34. Does it mean I *discovered* this (rather unuseful) property:

$$\mathbb{A} \setminus \mathbb{B} = \mathbb{A} \setminus (\mathbb{A} \cap \mathbb{B})$$

Or, perhaps a slightly more useful property[a]:

$$\begin{aligned} &\text{If} \\ &\quad \mathbb{A} = (\mathbb{D}_1 \cup (\mathbb{A} \cap \mathbb{B}), \&(\mathbb{D}_1 \cap (\mathbb{A} \cap \mathbb{B}) = \emptyset) \\ &\quad \mathbb{B} = (\mathbb{D}_2 \cup (\mathbb{A} \cap \mathbb{B}), \&(\mathbb{D}_2 \cap (\mathbb{A} \cap \mathbb{B}) = \emptyset) \\ &\text{Then} \\ &\quad \mathbb{A} \setminus \mathbb{B} = \mathbb{D}_1 \qquad (1.18) \\ &\text{And} \\ &\quad \mathbb{B} \setminus \mathbb{A} = \mathbb{D}_2 \qquad (1.19) \end{aligned}$$

?

Looks like that!

[a] Thanks again to Deepa Joshi for a discussion during which an egregious error in a previous formulation of this property was revealed.

Exercise 1.8.19. *Prove that for any three sets* $\mathbb{A}, \mathbb{B}, \mathbb{C}$*,*

$$((\mathbb{A} \setminus \mathbb{B}) \cup (\mathbb{B} \setminus \mathbb{A})) \cap \mathbb{C} = ((\mathbb{A} \cap \mathbb{B}) \cup (\mathbb{B} \cap \mathbb{C})) \setminus (\mathbb{A} \cap \mathbb{B} \cap \mathbb{C}) \qquad (1.20)$$

Solution.

$$((A \setminus B) \cup (B \setminus A)) \cap C \qquad ((A \cap C) \cup (B \cap C)) \setminus (A \cap B \cap C)$$

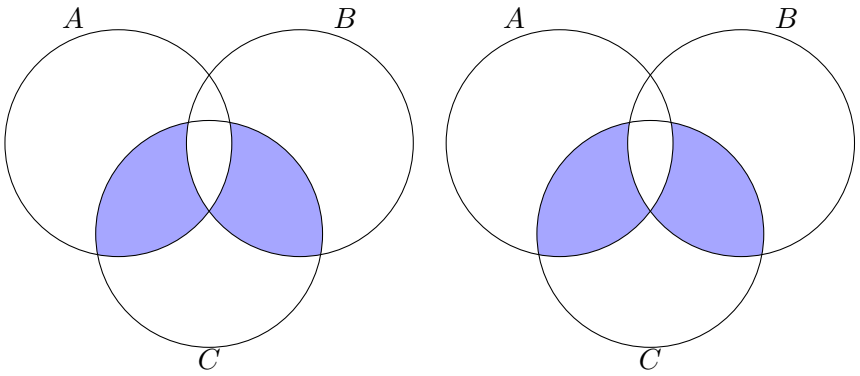


Figure 1.6: Figures drawn using Tikz by ChatGPT; Thanks!

Reflection 35. Is there another way to prove?

Exercise 1.8.20. *Let $\mathbb{A}$ be a set and let $\binom{\mathbb{A}}{2}$ denote the set of all 2-element subsets of $\mathbb{A}$. Which of the following statements is true?*

(a) $\binom{\mathbb{A}\cup\mathbb{B}}{2} = \binom{\mathbb{A}}{2} \cup \binom{\mathbb{B}}{2}$

(b) $\binom{\mathbb{A}\cup\mathbb{B}}{2} \supseteq \binom{\mathbb{A}}{2} \cup \binom{\mathbb{B}}{2}$

(c) $\binom{\mathbb{A}\cap\mathbb{B}}{2} = \binom{\mathbb{A}}{2} \cap \binom{\mathbb{B}}{2}$

(d) $\binom{\mathbb{A}\cap\mathbb{B}}{2} \subseteq \binom{\mathbb{A}}{2} \cap \binom{\mathbb{B}}{2}$

Solution. (a) is not true because there are elements in $\binom{\mathbb{A}\cup\mathbb{B}}{2}$ that are not in $\binom{\mathbb{A}}{2} \cup \binom{\mathbb{B}}{2}$ (e.g. a 2-subset whose one element is exclusively in $\mathbb{A}$ and another exclusively in $\mathbb{B}$ is in $\binom{\mathbb{A}\cup\mathbb{B}}{2}$, but not in $\binom{\mathbb{A}}{2} \cup \binom{\mathbb{B}}{2}$). Consequently, every *pair of elements* exclusively in $\mathbb{A}$ or exclusively in $\mathbb{B}$ is also in the set $\binom{\mathbb{A}\cup\mathbb{B}}{2}$, which means that $\binom{\mathbb{A}\cup\mathbb{B}}{2}$ is a superset of $\binom{\mathbb{A}}{2} \cup \binom{\mathbb{B}}{2}$ (the equality holds when $\mathbb{A}$ and $\mathbb{B}$ are the same set). Therefore, (b) is true.

Pairs of elements common to $\mathbb{A}$ and $\mathbb{B}$ are the only ones that are the members of both $\binom{\mathbb{A}\cap\mathbb{B}}{2}$ and $\binom{\mathbb{A}}{2} \cap \binom{\mathbb{B}}{2}$. Therefore, (c) and (d) are also true.

Exercise 1.8.21. *Let $\mathbb{B}$ be a subset of $\mathbb{A}$, $|\mathbb{A}| = n$, and $|\mathbb{B}| = k$. What is the number of all subsets of A whose intersection with $\mathbb{B}$ has 1 element?*

Solution. If we represent the sets by circles, since $\mathbb{B} \subset \mathbb{A}$, the circle representing $\mathbb{B}$ is *inside (or coincides with)* the circle representing $\mathbb{A}$. To satisfy our condition, exactly one element of any subset of $\mathbb{A}$ must be in $\mathbb{B}$.

How many 1-subsets of $\mathbb{A}$ share 1 element with $\mathbb{B}$? Clearly, each element of $\mathbb{B}$ (which can be chosen $k \times \binom{n-k}{0} = k$ ways, since $|\mathbb{B}| = k$ and we choose no element from the remaining $n-k$).

How many 2-subsets of $\mathbb{A}$ have that property? The subsets for which one element is in $\mathbb{B}$ and one in the set $\mathbb{A} \setminus \mathbb{B}$. The number of 2-subsets is, therefore, equal to the number of ways to choose the two elements together, that is, $k \times (n-k)$ or $k \times \binom{n-k}{1}$.

The 3-subsets satisfying that condition are the ones for which one element is in $\mathbb{B}$ and two in the set $\mathbb{A} \setminus \mathbb{B}$. That number is $k \times \binom{n-k}{2}$.

This continues up to the $(n-1)$-subsets of $\mathbb{A}$. Therefore, the total number of subsets of $\mathbb{A}$ with one element in $\mathbb{B}$, s_{1B} is

$$
\begin{aligned}
s_{1B} &= k \times \binom{n-k}{0} + k \times \binom{n-k}{1} + k \times \binom{n-k}{2} + \cdots + k \times \binom{n-k}{n-k} \\
&\quad \text{(since there are } n-k \text{ elements in } \mathbb{A} \setminus \mathbb{B}\text{)} \\
&= k \times \sum_{i=0}^{n-k} \binom{n-k}{i} \\
&= k \times 2^{n-k}
\end{aligned}
$$

Reflection 36 (An Alternative Solution). Deepa Joshi made a compelling argument: Since $|\mathbb{A}| = n$, $|\mathbb{B}| = k$, and $\mathbb{B} \subset \mathbb{A}$, the set $\mathbb{A} \setminus \mathbb{B}$ has $n - k$ elements. If we add an element of $\mathbb{B}$ (which is also an element of $\mathbb{A}$) to each subset of $\mathbb{A} \setminus \mathbb{B}$, we get a set whose intersection with $\mathbb{B}$ has just one element (namely, the element that was added).

Since the number of subsets of $\mathbb{A} \setminus \mathbb{B}$ is 2^{n-k} and there are k ways to choose one element of $\mathbb{B}$, by the multiplication principle (definition [1]) , the number of ways of making the two choices together is

$$k \times 2^{n-k}$$

(This argument feels elegant!)

Exercise 1.8.22. *Compute the binary form of* 25 *and* 35*, and compute their sum in the binary notation. Check the results against adding* 25 *and* 35 *in the usual decimal notation and then converting it to binary.*

Solution. **TBD**

Exercise 1.8.23. *Prove that every positive integer can be written as the sum of different powers of* 2*. Also prove that for a given number, there is only one way to do so.*

Solution.

Reflection 37. There are several ways to prove this. Since we are asked to prove that as a review exercise on a chapter on discrete mathematics, perhaps we should make a combinatorial argument.

Proof. We *write* a natural number (a) in the *place-value system* of any base greater than 1 (b) by repeatedly dividing a by b. From the division algorithm, it follows that for any two integers a, b ($b \neq 0$), there exist two integers q, r, called quotient and remainder respectively, such that

$$a = bq + r \wedge 0 \leq r < b$$

On the first division, we get $a = bq_1 + r_1$ $(0 \leq r_1 < b)$. Then we write r_1.

If q_1 is larger than b, we perform another division to get $q_1 = bq_2 + r_2$, or $a = b(bq_2 + r_2) + r_1 = b^2q_2 + br_2 + r_1$ $(0 \leq r_1, r_2 < b)$. Then we write r_2 to the left of r_1. So far, we have written two 'digits' both less than b: r_2r_1 and we need to 'process the remains' in q_2.

It follows that the quotient q_i become smaller with each division. We stop the process when a certain q_i becomes less than b, after which all q_i are 0. This is the digit we finally write. The *value* of our number is

$$a = b^{n-1}q_n + b^{n-2}r_{n-1} + \cdots + b^2r_3 + br_2 + r_1 \tag{1.21}$$

and we *write* it as[10]:

$$q_nr_{n-1} \ldots r_3r_2r_1 \tag{1.22}$$

Each of the digits in equation [1.22] is less than b. This means that when $b = 2$, each of the digits in equation [1.22] is either 0 or 1. It then follows that the value of the number given by equation [1.21] is a sum of powers of 2.

From the multiplication principle (definition [1]), it follows that if we have n binary digits (bits), we can choose them together 2^n *distinct ways*: each representation (a binary n-string) differs from every other representation in at least one bit. Thus, the set of 2^n binary representations and the set of numbers each binary representation represents form a bijection (a 1:1 mapping). This proves that every binary representation uniquely represents an integer. ■

[10] In practice, we usually start writing from the big end, i.e., q_n in this case.

Exercise 1.8.24. *How many bits does* 10^{100} *have if written in base* 2*?*

Solution. If we have n *bits* to write the *binary* representation of a number, the minimum value of that number occurs when the most significant bit is 1 and other bits are zero:

$$\underbrace{1\,000\ldots0}_{n-1 \text{ bits}}: \quad 1\times 2^{n-1} + \underbrace{0\times 2^{n-2} + \cdots + 0\times 2^1 + 0\times 2^0}_{n-1 \text{ bits}} = 2^{n-1}$$

and the maximum value occurs when *all* bits are 1:

$$\underbrace{1111\ldots1}_{n \text{ bits}}: \quad \underbrace{1\times 2^{n-1} + 1\times 2^{n-2} + \cdots + 1\times 2^1 + 1\times 2^0}_{n \text{ bits}} = 2^n - 1$$

Therefore, the number that we want to represent as a binary string, 10^{100}, must be between these two limits:

$$2^{n-1} \leq 10^{100} \leq (2^n - 1)$$

or, equivalently,

$$2^{n-1} \leq 10^{100} < 2^n$$

Taking logarithms to the base 2, we get

$$\begin{aligned} n - 1 &\leq 100 \times \log_2 10 < n \\ n - 1 &\leq 100 \times 3.321928094887362 < n \\ n - 1 &\leq 332.1928094887362 < n \\ \therefore n &= 333 \end{aligned}$$

Reflection 38. The number, 1 followed by 100 zeros, 10^{100}, has a special name: Googol. Will there be enough space to write here the number of digits (and bits) that the number $10^{10^{100}}$ (Googolplex) has?

Exercise 1.8.25. *Starting from Washington, DC, how many ways can you visit* 5 *of the* 50 *state capitals and return to Washington?*

Solution. There are various ways of prove this, but for brevity, the answer[11] is $50 \times 49 \times 48 \times 47 \times 46$.

Exercise 1.8.26. *Find the number of all 20-digit integers in which no two consecutive digits are the same.*

Solution. There are 9 choices $(1, 2, 3, \ldots,$ or $, 9)$ for the most significant digit of an n-digit integer. For every other digit, however, there are 10 choices $(0, 1, 2, \ldots,$ or $, 9)$. In a number C that has no consecutive pair, after we choose a first digit (say 3), we may not choose 3 for the second digit (but we may choose 3 for any subsequent digit). That leaves only 9 choices for the second digit. The number of choices for subsequent digits also remains 9. Thus, the digits of C can be chosen together

$$\underbrace{9\times 9\times 9\times \cdots \times 9}_{n \text{ digits}} = 9^n$$

ways.

Therefore, the number of 20-digit numbers with no consecutive pair of the same digit is 9^{20}.

Reflection 39 (Two Ways)**.** The above argument sounds correct. However, I am a bit apprehensive. I feel the need of solving this problem another way. Addressing this apprehension is why Lovász et.al. want us to solve combinatorial problems two different ways.

Solving it another way is somewhat challenging. Let's think of 4-digit decimal numbers. There are $9 \times 10 \times 10 \times 10 = 9000$ of them. How many of them have at least one consecutive pair of a digit (e.g. 1128)? Directly counting them by fixing a same-digit-consecutive-pair in various positions and counting

[11] Note: Washington, DC, is not one of the 50 state capitals; had it been, the answer would be different.

choices for the remaining digits feels tricky in the general case. There is an overcounting trap laid by numbers like 1111 or 2244.

Will an inductive argument help? For example, there are 9 numbers with a same-digit-consecutive-pair in 2-digit numbers:11, 22, 33, 44, 55, 66, 77, 88, 99. How can we use that fact to ascertain the 3-digit numbers with a same-digit-consecutive-pair? In the range 100-199, there are 100, (110, 111, 112, . . . , 119, and (122, 133, . . . , 188, 199), that is, $1+10+8=19$. This holds for every hundred up to 900-999. That gives us $19 \times 9 = 171$ such numbers for the 3-digit case. However, I don't see a pattern yet.

Exercise 1.8.27. *Alice has* 10 *balls (all different). First, she splits them into two piles; then she picks one of the piles with at least two elements, and splits it into two; she repeats this until each pile has only one element.*

(a) How many steps does this take?

(b) Show that the number of different ways in which she can carry out this procedure is

$$\binom{10}{2} \times \binom{9}{2} \times \binom{8}{2} \times \cdots \times \binom{3}{2} \times \binom{2}{2}$$

[Hint: Imagine the procedure backward.]

Solution.

(a) If there are 2 balls, the only step is to split them into two parts of 1 ball each and the procedure ends. Any number of balls $n > 2$, can be split into two parts with one part of 2 balls (which, if chosen, can be split in 2 parts) and one of $n-2$. Thus, the minimum number of steps is 2.

If, instead of splitting n balls such that there are 2 in one part, we split n balls into two parts with 1 in one part and $n-1$ in the other, we must only choose the part that has $n-1$ balls. This procedure, if continued like this, will take the maximum number of steps, $n-1$.

Therefore, the number of steps is any integer between $[2, n-1]$. The procedure can be 'short-circuited' by splitting into two parts with one part of 2 balls and choosing that part next.

(b) **TBD** (I need to think a bit more; I will revisit this.)

Exercise 1.8.28. *You want to send postcards to* 12 *friends. In the shop there are only* 3 *kinds of postcards. In how many ways can you send the postcards, if*

(a) there is a large number of each kind of postcards, and you want to send one card to each friend;

(b) there is a large number of each kind of postcards, and you are willing to send one or more postcards to each friend (but no one should get two identical cards);

(c) the shop has only 4 *of each kind of postcards, and you want to send one card to each friend?*

Solution.

(a) 3^{12}.

Reflection 40 (All Textbook Combinatorial Problems are Word-problems)**.** I was a bit confused by this problem and actually worked out a solution that equals

$$\binom{12}{0} \times 2^{12} + \binom{12}{1} \times 2^{11} + \binom{12}{2} \times 2^{10} + \cdots + \binom{12}{11} \times 2^{1} + \binom{12}{12} \times 1$$

, which is what we get on evaluating the binomial expansion

$$(2+1)^{12} = 3^{12}$$

I considered three kinds of postcards: R, G, and B. Then, I send no R card but all 12 cards are either G or B kind:

a) R:0 – can be done only one way.

1) G:0, B:12. There is $\binom{12}{0} = 1$ way of doing this: every friend gets a B card.

2) G:1, B:11. There are $\binom{12}{1} = 12$ ways of doing this.

3) G:2, B:10. There are $\binom{12}{2}$ ways of doing this.

$\vdots$

12) G:12, B:0. There is $\binom{12}{12} = 1$ way of doing this: every friend gets a G card.

The sum of these ways (when I decide to send no R card) is

$$\binom{12}{0} + \binom{12}{1} + \cdots + \binom{12}{11} + \binom{12}{12} = 2^{12}$$

b) R:1 – send an R card to one friend who can be chosen $\binom{12}{1}$ ways. The sum of these ways (when I decide to send one R card) is

$$12 \times (\binom{11}{0} + \binom{11}{1} + \cdots + \binom{11}{10}\binom{11}{11}) = 12 \times 2^{11}$$

$\vdots$

l) R:12 – send every friend an R card; this can be done one way.

I soon realized that I solved the problem the hard way.

Deepa Joshi heard the problem and said, "Well, it's 3^{12}, because every friend can receive an R, G, or B card and there are 12 friends. Use the multiplication principle!" I wonder why it did not occur to me!

(b) I took time to realize[12] that I should focus on the choices for one friend. Since there are many cards of three kinds and a friend can get one or more cards of unique kind, the choices for each friend are: R (1), G (1), B (1), R, G (2), G, B (2), B, R (2), R, G, B (3). These are 7. Therefore, the number of ways of sending that many postcards, from multiplication principle, is

$$7^{12}$$

(c) I put my heart into this one and decided to solve it two different ways. I am glad I was able to get the same answer! I wonder where the intuition of solving problems comes from!

1) There are 3 kinds of postcard and 4 postcards of a kind. That makes $3 \times 4 = 12$ postcards. There are 12 friends and 12 cards. That makes 12! friend-postcard assignments possible. However, every R card is indistinguishable from every other R card. Thus, if we, only for brevity, mark the R cards as R1, R2, R3, R4 (and G and B cards similarly), then the two assignments for friends numbered 1 through 12: 1) **R1, R2**, R3, R4, B1, B2, B3, B4, G1, G2, G3, G4, and (for example), and 2) **R2, R1**, R3, R4, B1, B2, B3, B4, G1, G2, G3, G4 are identical and must not be counted as different.

In other words, the 4! R assignments, 4! G assignments, and 4! B assignments result in overcounting. By addressing this overcounting, the number of ways to send postcards (with the new constraint) is

$$\frac{12!}{4! \times 4! \times 4!}$$

[12] Deepa Joshi helped again!

2) We can devise a single assignment of 12 cards to 12 friends by another strategy: Choose any 4 friends and assign a card of any kind two them. Then, assign 4 cards of another kind to any 4 of the remaining 8. Then, the remaining 4 friends get a card of the remaining kind. Together, these assignments can be done (by using the multiplication principle)

$$\binom{12}{4} \times \binom{8}{4} \times \binom{4}{4} = \frac{12!}{4! \times \cancel{8!}} \times \frac{\cancel{8!}}{4! \times 4!} \times 1 = \frac{12!}{4! \times 4! \times 4!}$$

ways.

Exercise 1.8.29. *What is the number of ways to color n objects with 3 colors if every color must be used at least once?*

Solution. We state that an object is colored entirely in one color. If $n < 3$, there's no way of doing this because at least one color will be unused.

For $n \geq 3$, we can choose any 3 objects and color them in 3 colors such that every color is used at least once $\binom{n}{3} \times 3!$ ways. Once that constraint has been satisfied, the remaining $n-3$ objects can be colored such that every object is colored in any one of the 3 colors. Therefore, the total number of ways to do this is

$$\binom{n}{3} \times 3! \times 3^{n-3}$$

Exercise 1.8.30. *Draw a tree for Alice's solution of enumerating the number of ways 6 people can play chess, and explain Alice's argument using the tree.*

Solution.

TBD

Exercise 1.8.31. *How many different words can you get by rearranging the letters in the word MATHEMATICS?*

Solution. We make 11-letter words by rearranging in 11 positions the 11 letters of the word MATHEMATICS, which has 2 M's, 2 A's, and 2 T's. Therefore, we must account for overcounting. The number of ways to achieve this is

$$\frac{11!}{2! \times 2! \times 2!}$$

Exercise 1.8.32. *Find all positive integers a, b, and c for which*

$$\binom{a}{b} \times \binom{b}{c} = 2\binom{a}{c}$$

Solution. **TBD**

Exercise 1.8.33. *Prove that*

$$\binom{n}{k} = \binom{n-2}{k} + 2\binom{n-2}{k-1} + \binom{n-2}{k-2} \tag{1.23}$$

Solution. We use equation [1.10] twice, once on $\binom{n}{k}$ and once on $\binom{n-1}{k-1}$, and what needs to be proved follows.

Exercise 1.8.34. *20 persons are sitting around a table. How many ways can we choose 3 persons, no two of whom are neighbors?*

Solution.

Reflection 41. I want to solve it two different ways! The first thought was an indirect way.

(a) For simplicity, we number them $1, 2, 3, \ldots, n-1, n$ in a clockwise manner. There are $\binom{n}{3}$ ways to choose 3 (let's call the selection a triplet) from n people sitting around a table. In a triplet like $\{1, 2, 3\}$ or $\{2, 3, 14\}$ at least one pair of neighbors is chosen. We do not want to choose such triplets. How many such 'neighbor-triplets' are there?

Triplets with the starting neighbor pair $1, 2$ are: $\{1, 2, 3), \{1, 2, 4\}, \{1, 2, 5\}, dots, \{1, 2, n\}$. These are $n - 3 + 1 = n - 2$.

Triplets with the starting neighbor pair $2, 3$ are: $\{2, 3, 4\}, \{2, 3, 5\}, \ldots, \{2, 3, n\}$, but not $\{2, 3, 1\}$ because we already covered that in $\{1, 2, 3\}$. These are $(n - 3)$. The number of such triplets is the same for every triplet up to the triplet whose starting neighbor pair is $n - 2, n - 1$.

The last starting neighbor pair, $n - 1, n$ has two fewer, that is $n - 4$, because both triplets $\{n - 1, n, 1\}$ and $\{n, 1, 2\}$ are already counted.

Thus, the number of overcounted triplets is

$$(n-2) \times 1 + (n-3) \times (n-2) + (n-4) \times 1 = 2n\cancel{-6} + n^2 - 5n\cancel{+6} = n^2 - 3n$$

Therefore, the number of triplets without any neighbor is

$$\binom{n}{3} - (n^2 - 3n) = \frac{n \times (n-1) \times (n-2)}{6} - n^2 + 3n = \frac{n \times (n-4) \times (n-5)}{6}$$

(b) Another way to calculate the number of neighbor-triplets is to consider the selection with a slight twist. A neighbor-triplet has a neighbor pair and a third person (who may be a neighbor of one of the members of the pair). With n people, the number of neighbor-pairs ($\{1, 2\}, \{2, 3\}, \ldots, \{n-1, n\}, \{n, 1\}$) is n; we order the pairs in a clockwise manner for the ease of counting.

Once a pair is chosen, to avoid overcounting, we pick the third member of the triplet only in one (clockwise or counter clockwise) direction. Thus, for the pair $\{3, 4\}$, we pick one person from $\{5, 6, 7, \ldots, n-1, n, 1\}$, for which we have $n - 3$ choices; we skip 2, because we have already considered the triplet $\{3, 4, 2\}$ while dealing with the pair $\{2, 3\}$ as $\{2, 3, 4\}$. From the multiplication principle, it follows that the ways to choose such triplets is $n \times (n - 3) = n^2 - 3n$.

Therefore, the number of triplets without any neighbor is

$$\binom{n}{3} - (n^2 - 3n)$$

Reflection 42 (Finishing up, Checking Answers!)**.** Phew!

This has been a great ride. I need to verify my answers, but completing the attempt to solve these exercises does feel like an achievement of sorts and small achievements like this satisfy me.

Should I go ahead with the next chapter for that means progress? Yes, but not before revising the ideas in this chapter.

I revised the ideas, will check the answers!

TBD

Chapter 2

Combinatorial Tools

Reflection 43. Progress feels good. I started with this book on 20 August 2026. Today is 20 October 2026. Through other commitments, distractions, shifting desires, (temporary) loss or lack of creative argumentation and/or memory, I have reached the end of the first chapter. I hope I continue this momentum. Will it get easier, easier meaning less time-consuming, more intuitive, and more rewarding? Will the knowledge stay with me? Perhaps. I stay guided by what Alan Kay told Bret Victor [2]:

> *I think the trick with knowledge is to "acquire it, and forget all except the perfume" – because it is noisy and sometimes drowns out one's own "brain voices". The perfume part is important because it will help find the knowledge again to help get to the destinations the inner urges pick.*
>
> — Alan Kay
> [2]

I am not exactly sure what my 'inner urges' to study discrete mathematics are, but for now, I want to do the ground work to enjoy playing and solving games and puzzles and appreciate Knuth's Volume 4A [3].

2.1 Induction

Definition 13 (Principle of Mathematical Induction)**.** Suppose that we want to prove a property of positive integers. Also suppose that we can prove two facts:

a) 1 has the property, and

b) Whenever $n; n > 1$ has the property, then $n + 1$ also has that property[1]

Then, every natural number has that property.

Exercise 2.1.1. *Prove, using induction but also without it, that $n(n+1)$ is an even number for every nonnegative integer n.*

Reflection 44. Only for practice.

[1] For some reason, the book favors: "Whenever $n - 1; n > 1$ has the property, then n also has that property;" we'll stick to $P(n) \implies P(n+1)$, rather than $P(n-1) \implies P(n)$, however. Clearly, it is a question of preference.

Solution.

Proof. We use mathematical induction. Let $P(n)$ be the proposition that $n(n+1)$ is even $\forall n > 0$.

Base case. We prove that $P(n)$ is true for $n = 1$. That is, $P(1)$ is true.

By definition, $P(1) = 1(1+1) = 1 \cdot 2 = 2$. Since 2 is even, $n(n+1)$ is even and $P(1)$ is true.

Inductive hypothesis. Let $P(n)$ be true for some $n = k$, that is, $k(k+1)$ is even. Therefore,

$$k(k+1) = 2m \mid m \in \mathbb{N} \tag{2.1}$$

We will show that $P(k+1) = (k+1)(k+2)$ is even.

$$\begin{aligned}(k+1)(k+2) &= k^2 + 2k + k + 2 \\ &= k^2 + k + 2k + 2 \\ &= 2m + 2(k+1) \text{ (from equation [2.1])} \\ &= 2(m + k + 1) \text{ (which is even, by definition)}\end{aligned}$$

Therefore, $P(k+1)$ is even. The property holds for all $n \in \mathbb{N}$. ■

Proof. We use contradiction.

Let $n(n+1)$ be odd. Since the product of two numbers is odd, they must be odd. Thus, n and $n+1$ are both odd. The difference between them is $n + 1 - n = 1$. That is a contradiction because the difference between any two odd numbers must be even.

Therefore, $n(n+1)$ is even. ■

Exercise 2.1.2. *Prove by induction that the sum of the first n positive integers is $\frac{n(n+1)}{2}$.*

Solution. It's fairly straightforward to realize that

$$\frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$$

, which proves the result.

Exercise 2.1.3. *Observe that $n(n+1)/2$ is the number of handshakes among $n+1$ people.*

Suppose that everyone counts only handshakes with people older than him/her (pretty snobbish, isn't it?). Who will count the largest number of handshakes? How many people count 6 handshakes? (We assume that no two people have exactly the same age.)

Give a proof of the result of exercise [2.1.2], based on your answers to these questions.

Solution. $n+1$ people convened. Everyone shook hands with everyone else. After they are done, an observer asks each person, "How many times did *you* shake hands?". Since each of the $n+1$ people shook hands with n other people, everyone says, "n times." The observer concludes that the combined number of 'shaking hands' of $n+1$ people, T, is $n(n+1)$.

However, is T the number of 'handshakes'? No! Since a handshake comprises two shaking hands, the observer must have counted each handshake twice. Therefore, the number of handshakes among $n+1$ people is $\frac{T}{2} = \frac{n(n+1)}{2}$.

Reflection 45. I got distracted a bit at this point. I knew I was close to seeing the connection between this and the previous problem, but there was a gap. [More time elapsed ...] I think I see it now ...

Counting T and then halving it is equivalent to imposing an order on counting. Here is a way to impose the order after ensuring that each person has a unique age.:

1. A person, p1, arrives at the scene and sits. There's no handshake yet.
2. An older person, p2, arrives. p1 gets up in reverence and shakes hands with p2. Both sit. The number of handshakes is $1 = \frac{1\times 2}{2} = 1$.
3. A still older person, p3, arrives. p1 and p2 get up in reverence and shake hands with p3 in order. The number of handshakes is now $1 + 2 = \frac{2\times 3}{2} = 3 = 1 + 2$.
4. A still older person, p4, arrives. p1, p2, and p3 get up in reverence and shake hands with p4 in order. The number of handshakes is now $3 + 3 = \frac{3\times 4}{2} = 6 = 1 + 2 + 3$.
5. A still older person, p5, arrives. p1, p2, p3, p4 get up in reverence and shake hands with p5 in order. The number of handshakes is now $6 + 4 = \frac{4\times 5}{2} = 10 = 1 + 2 + 3 + 4$.
6. The pattern continues. After p(n+1) has arrived, we have $\frac{n(n+1)}{2} = 1 + 2 + 3 + \cdots + n$ handshakes.

Exercise 2.1.4. *Give a proof of exercise [2.1.2], based on Figure [2.1].*

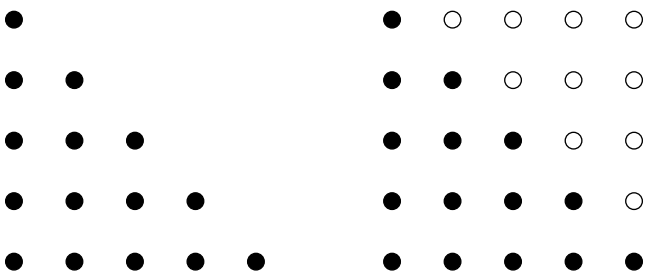


Figure 2.1: The sum of the first n integers (Tikz code by ChatGPT; Thanks!)

Solution. This is the Gauss trick.

Exercise 2.1.5. *Prove the following identity:*

$$1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + (n-1) \cdot n = \frac{(n-1) \cdot n \cdot (n+1)}{3} \tag{2.2}$$

Solution.

Reflection 46. Discovering a closed-form formula for an expression such as the left-hand side of the identity [2.2] (of course, without using the mathematical induction) is tricky; proving an established truth using induction is easier. Perhaps that is why Pólya[a] called induction a 'diabolical technique' of proving something.

I will try to discover this identity.

[a] I vaguely remember reading it attributed to Pólya.

Proof. We use mathematical induction to prove that the proposition

$$P(n) : 0 \cdot 1 + 1 \cdot 2 + 2 \cdot 3 + \cdots + (n-1) \cdot n = \frac{(n-1) \cdot n \cdot (n+1)}{3}$$

is true $\forall n \in \mathbb{N}$.

It is straightforward to see that

$$\frac{(n-1) \cdot \boldsymbol{n} \cdot (n+1)}{3} + n \cdot \boldsymbol{(n+1)} = \frac{(n-1) \cdot n \cdot (n+1) + 3n \cdot (n+1)}{3} = \frac{n \cdot \boldsymbol{(n+1)} \cdot (n+2)}{3}$$

, from which the proof follows. ■

Reflection 47 (A Slightly Less Diabolical Way)**.** We can easily deduce the following from the left-hand side of equation [2.2] (note the indices of the sum; they are from 0 to $n-1$):

$$1\cdot 2+2\cdot 3+3\cdot 4+\cdots+(n-1)\cdot n=\sum_{i=1}^{n-1} i(i+1)=\sum_{i=1}^{n-1}(i^2+i)$$

$$=\frac{(n-1)n(2n-1)}{6}+\frac{(n-1)n}{2}=\frac{(n-1)n\cdot 2(n+1)}{6}=\frac{(n-1)n(n+1)}{3}$$

, which was to be demonstrated.

If, on the other hand, the left-hand side were:

$$1\cdot 2+2\cdot 3+3\cdot 4+\cdots+\boldsymbol{n\cdot(n+1)}$$

, the closed-form formula would be:

$$\sum_{i=1}^{\boldsymbol{n}}(i^2+i)=\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}=\frac{\boldsymbol{n(n+1)(n+2)}}{\boldsymbol{3}}$$

Reflection 48 (Using a Technique I Learned Previously)**.** I thought of the previous exercise (2.1.5) more generally. It is a finite sum of products of two numbers which are in arithmetic progression:

$$S_n=a_1\cdot b_1+a_2\cdot b_2+\cdots+a_n\cdot b_n \mid a_1,a_2,\ldots,a_n \text{ and } b_1,b_2,\ldots,b_n \text{are in A.P.}$$

Will we be able to get a generalized result of the sum in terms of a_1, b_1, n, d_a, d_b?

I haven't worked this out on paper yet, but I believe we might be able to generalize! First, let me write down a few terms of our sum for various values of n. The leading $+$ in the table cells means we add a new product term to the sum for a previous value of n.

1	2	3	4	5		n
1·2 2	+2·3 8	+3·4 20	+4·5 40	+5·6 70	...	$+(n-1)\cdot n$

Table 2.1: The First Few Terms

Polynomials, one of the most familiar algebraic constructs, can help! Polynomials in a single variable are real-valued functions ($f:\mathbb{R}\to\mathbb{R}$), however, in this case we can look only at the values of polynomials for integral values of its variable (treating the function as $f:\mathbb{N}\to\mathbb{R}$).

Let n be the variable of the polynomial, which assumes only integral values. Let's assume that the sum S_n is a polynomial[a] in n.

Here are the terms of S_n and various finite differences for various n (we assume a $\cdot$, but don't write it in expressions involving products):

n	Sum	Δ_1
1	$S_1 = a_1b_1$	-
2	$S_2 = S_1 + (a_1 + d_a)(b_1 + d_b)$	$(a_1 + d_a)(b_1 + d_b)$
3	$S_3 = S_2 + (a_1 + 2d_a)(b_1 + 2d_b)$	$(a_1 + 2d_a)(b_1 + 2d_b)$
4	$S_4 = S_3 + (a_1 + 3d_a)(b_1 + 3d_b)$	$(a_1 + 3d_a)(b_1 + 3d_b)$
5	$S_5 = S_4 + (a_1 + 4d_a)(b_1 + 4d_b)$	$(a_1 + 4d_a)(b_1 + 4d_b)$
$\vdots$	$\vdots$	$\vdots$
$n-1$	$S_{n-1}=S_{n-2}+(a_1+(n-2)d_a)(b_1+(n-2)d_b)$	$(a_1+(n-2)d_a)(b_1+(n-2)d_b)$
n	$S_n=S_{n-1}+(a_1+(n-1)d_a)(b_1+(n-1)d_b)$	$(a_1+(n-1)d_a)(b_1+(n-1)d_b)$

Table 2.2: Terms of the Sum and First Finite Differences

n	Δ_2	Δ_3
1	-	-
2	-	-
3	$a_1d_b + b_1d_a + 3d_ad_b$	-
4	$a_1d_b + b_1d_a + 5d_ad_b$	$2d_ad_b$
5	$a_1d_b + b_1d_a + 7d_ad_b$	$2d_ad_b$
$\vdots$	$\vdots$	$\vdots$
$n-1$	$a_1d_b + b_1d_a + (2n-5)d_ad_b$	$2d_ad_b$
n	$a_1d_b + b_1d_a + (2n-3)d_ad_b$	$2d_ad_b$

Table 2.3: Terms of Second and Third Finite Differences

The technique is called 'finite differences.' We get a sequence of terms (which are partial sums), S_i. Then, $\Delta_i = S_i - S_{i-1} \forall i \mid 2 \leq i \leq n$ are called the first finite differences. The differences in first differences are called second finite differences (Δ_i^2), and so on.

According to a theorem, if the nth finite differences, Δ_n's, are the same, the degree of the polynomial generating the terms is also n (and vice versa).

Therefore, the polynomial in n whose coefficients we seek is:

$$c_0 + c_1n + c_2n^2 + c_3n^3 \mid c_3 \neq 0 \tag{2.3}$$

We find the coefficients by considering the terms of the sum we know. We only make one change for the ease of calculation: Make[b] the first term of the sum $0 \cdot 0$. :

n	Expression$c_0 + c_1n + c_2n^2 + c_3n^3$	Value
1	$c_0 + c$	0
2		$a_1 + b_1$
3		$a_1 + b_1 + (a_1 + d) \cdot (b_1 + d)$
4		

[a] We get a hint from exercise 2.1.5 anyway.

[b] This is only to adjust n so that it works for all cases. For example, in the given sum $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + (n-1) \cdot n$, with only one term (the first one) the sum is $1 \cdot 2 = 2$, but if we substitute $n = 1$ in $\frac{(n-1)n(n+1)}{3}$, we get 0. We work around this problem, by making the first term $0 \cdot 0 = 0$.

Exercise 2.1.6. *Use the method of the young Gauss to give a third proof of the formula in exercise 2.1.2.*

Solution. The generalization of the concrete example (when $n = 1000$) is easy.

Exercise 2.1.7. *How would the young Gauss prove the formula*

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2 \tag{2.4}$$

for the sum of the first n odd numbers?

Solution.

Proof. We prove it the Gauss way.

$$\begin{array}{rrr} S = 1 & +3 + \ldots & +(2n-1) \\ S = (2n-1) & +(2n-3) + \ldots & +1 \\ 2S = 2n & +2n + \ldots & +2n \end{array}$$

There are n addends in $2S$, each of which is $2n$. Therefore,

$$2S = 2n \times n$$

$$\therefore S = n^2$$

■

Exercise 2.1.8. *Prove that the sum of the first n squares $1 + 4 + 9 + \cdots + n^2$ is $\frac{n(n+1)(2n+1)}{6}$.*

Reflection 49. This is a classic problem in number theory and the technique of finite differences can be applied to it. A blind application of the young Gauss's insight does not work immediately. There are several subproblems here. Why doesn't Gauss's technique work as is? Is the product $n(n+1)(2n+1)$ divisible by $6 \forall n \in \mathbb{N}$ (it must be, because, otherwise the sum of integers won't be an integer)?

Solution. It's pretty straightforward to prove using induction and finite differences.

Exercise 2.1.9. *Prove that the sum of the first n powers of 2 (starting with $1 = 2^0$) is $2^n - 1$.*

Solution.

Reflection 50. Here, the technique of polynomials does not work because the sum is expected to be exponential in n.

Using induction, it's fairly easy to see that if

$$\sum_{i=0}^{n-1} 2^i = \underbrace{1 + 2 + 4 + \cdots + 2^{n-1}}_{n \text{ powers of } 2} = 2^n$$

, then

$$\mathbf{1 + 2 + 4 + \cdots + 2^{n-1}} + 2^n = \mathbf{2^n} + 2^n = 2 \times 2^n = 2^1 \times 2^n = 2^{n+1}$$

, and the proof follows from induction.

The book then explains an inductive argument that the number of subsets of an n-set, $\mathbb{S}$, is 2^n. The argument revolves around fixing an element a of the set and then considering the number of subsets (2^{n-1}) of the set $\mathbb{S} - \{a\}$.

Exercise 2.1.10. *Use induction to prove Theorem 1.5.1 (the number of strings of length n composed of k given elements is k^n) and Theorem 1.6.1 (the number of permutations of a set with n elements is $n!$).*

Solution. Inductive arguments are straightforward.

Exercise 2.1.11. *Use induction on n to prove the "handshake theorem" (the number of handshakes between n people is $\frac{n(n-1)}{2}$).*

Solution.

$$\underbrace{\frac{(n-1)(n-2)}{2}}_{\text{num of handshakes between n-1 people}} + \overbrace{(n-1)}^{\text{num of handshakes of nth person}} = \frac{n(n-1)}{2}$$

Exercise 2.1.12. *Read carefully the following induction proof:*

ASSERTION: $n(n+1)$ is an odd number for every n.

PROOF: Suppose that this is true for $n-1$ in place of n; we prove it for n, using the induction hypothesis. We have

$$n(n+1) = (n-1)n + 2n$$

Now here $(n-1)n$ is odd by the induction hypothesis and $2n$ is even. Hence $n(n+1)$ is the sum of an odd number and an even number and it is odd.

The assertion that we 'proved' is obviously wrong for $n = 10 : 10 \cdot 11 = 110$ is even.

What is wrong with the proof?

Solution.

The 'proof' does not prove the mundane base case! For an inductive proof to be valid, we must show that it holds for a base case like, for example, $n = 0$ or $n = 1$. For both these values, the product $n(n+1)$ fails to be odd. Therefore, the proof is bogus.

Exercise 2.1.13. *Read carefully the following induction proof:*

ASSERTION: If we have n lines in a (2-D?) plane, no two of which are parallel, then they all go through one point.

PROOF: The assertion is true for one line (and also for 2, since we have assumed that no two lines are parallel).

Suppose that it is true for any set of $n-1$ lines. We are going to prove that it is also true for n lines, using this induction hypothesis.

So consider a set $\mathbb{S} = \{a, b, c, d, \dots\}$ of n lines in the plane, no two of which are parallel. Delete the line c; then we are left with a set $\mathbb{S}'$ of $n-1$ lines, and obviously, no two of these are parallel. So we can apply the induction hypothesis and conclude that there is a point P such that all the lines in $\mathbb{S}'$ go through P. In particular, a and b go through P, and so P must be the point of intersection of a and b.

Now put c back and delete d, to get a set $\mathbb{S}''$ of $n-1$ lines. Just as above, we can use the induction hypothesis to conclude that these lines go through the same point P'; but just as above, P' must be the point of intersection of a and b. Thus $P' = P$. But then we see that c goes through P. The other lines also go through P (by the choice of P), and so all the n lines go through P.

But the assertion we 'proved' is clearly wrong; where is the error?

Solution.

Very clever! However, the 'proof' does not show if the line c passes through the point of intersection P of a, b: we temporarily delete c, and when it reappears we assume that it passes through P. If c did pass through P before it was deleted, the argument is valid because we are considering a specific case that all n lines are passing through P. But what about the legitimate case that c does not pass through P?

Reflection 51. That took some time. I was a bit annoyed by that problem!

Time to check answers, but I'll revisit that. **TBD**

2.2 Comparing and Estimating Numbers

> *It is nice to have formulas (closed-form) for certain numbers (for example, for the number $n!$ of permutations of n elements), but it is often more important to have a rough idea about how large these numbers are. For example, how many digits does $100!$ have?*
>
> *Let us start with simpler questions. Which is larger, n or $\binom{n}{2}$?*
>
> — PG. 30

The trick to answer that simpler question is to take the ratio: $\frac{\binom{n}{2}}{n} = \frac{\frac{n(n-1)}{2}}{n} = \frac{n-1}{2}$. Since this ratio increases arbitrarily as n increases, the value of $\binom{n}{2}$ becomes much more than n.

Exercise 2.2.1. *(a) Prove that*

$$2^n > \binom{n}{3} \text{ if } n \geq 3.$$

(b) Use (a) to prove that $\frac{2^n}{n^2}$ becomes arbitrarily large as n becomes large.

Solution.

(a) *Proof.* Consider the two sequences ($\forall n \in \mathbb{N} \mid n \geq 3$):

$$8, 16, 32, 64, 128, \ldots, \qquad 2^{n-1}, 2^n \tag{s1}$$

$$1,\ 4,\ 10, 20,\ \mathbf{35}, \ldots, \qquad \binom{n-1}{3}, \binom{n}{3} \tag{s2}$$

$$3,\ 4,\ \ 5,\ \ 6,\ \ \mathbf{7}, \ldots, \qquad n-1, n$$

In the sequence (s1), every term is double its predecessor: $a_{n+1} = 2 \times a_n \forall n \geq 3$. In the sequence (s2), however, the ratio is

$$\frac{\binom{n+1}{3}}{\binom{n}{3}} = \frac{\frac{(n+1)\cancel{n(n-1)}}{\cancel{3!}}}{\frac{\cancel{n(n-1)}(n-2)}{\cancel{3!}}} = \frac{n+1}{n-2}$$

It can be easily shown that $\frac{n+1}{n-2} < \mathbf{2}\ \forall \boldsymbol{n} > \mathbf{5}$. And we see that the first four terms (when $n = 3, 4, 5, 6$) of the (s1) are all greater than the corresponding terms of (s2). There is no $n \geq 3$ for which $\binom{n}{3}$ is greater than or equal to 2^n. Starting from 64, we keep doubling to get subsequent terms of (s1), whereas starting from 20, we multiply by a number less than 2 to get subsequent terms of (s2). Therefore, $2^n > \binom{n}{3}\ \forall n \geq 3$. ■

(b) *Proof.* For $\binom{n}{3} = \frac{n(n-1)(n-2)}{3}$ to be greater than n^2, $n > 8$. Since, from (a) it follows that 2^n is greater than $\binom{n}{3}$, which in turn is greater than $n^2\ \forall n > 8$, it follows that $2^n > n^2\ \forall n > 8$. ■

It is difficult to estimate the value of $n!$. One crude range for its value is clear:

$$1 \times \underbrace{2 \times 2 \times 2 \times \cdots \times 2}_{n-1 \text{ times}} \leq 1 \times 2 \times 3 \times \cdots \times n \leq 1 \times \underbrace{n \times n \times n \times \cdots \times n}_{n-1 \text{ times}}$$

i.e.,

$$2^{n-1} \leq n! \leq n^{n-1}$$

Exercise 2.2.2. *Use induction to prove that*

$$n! > 2^n \ \forall n \geq 4$$

Solution.

Proof. We use mathematical induction.

Base case: For $n = 4$, $n! > 2^n$. The base case does not hold for $n < 4$.

Inductive hypothesis: Let $P(n)$: $n! > 2^n$ be true for some $k > 4$. Then, $(k+1)! = (k+1)k! > (k+1)2^k$.

$$\therefore (k+1)! > \frac{k+1}{2} 2^{k+1}$$

But $\forall k > 4$,

$$\frac{k+1}{2} > \frac{4+1}{2} > 2 > 1$$
$$\therefore (k+1)! > 2^{k+1}$$

■

There is a formula that gives a very good approximation of $n!$ We state it without proof, since the proof (although not terribly difficult) needs calculus.

Theorem 2.2.1 (Stirling's Formula or Stirling's Approximation)**.**

$$n! \sim (\frac{n}{e})^n \sqrt{2\pi n}$$

Here $\pi = 3.1415926\ldots$ is the area of unit circle, $e = 2.71828\ldots$ is the base of the natural logarithm, and $\sim$ means approximate equality in the precise sense that

$$\lim_{n\to\infty} \frac{n!}{(\frac{n}{e})^n \sqrt{2\pi n}} = 1$$

(that is, the two quantities are asymptotic.)

Reflection 52. Stirling's formula (2.2.1) is amazing. How does one go about creating it? Should I check its Wikipedia page, or ask ChatGPT about it?

I better learn calculus well!

Even so, Stirling's Approximation is unlike usual closed-form formulas; you must find n^n in order to find $n!$. However, taking logarithms on both sides helps in certain cases. For instance,

$$\ln n! = n \ln n - n + O(\ln n)$$

We have seen before (section 1.4) that the number of digits in the decimal representation of any number is its logarithm to base 10 rounded up. We can utilize it to find the number of decimal digits in 100!. From the statement of Stirling's approximation for $n = 100$,

$$100! \approx (\frac{100}{e})^{100} \sqrt{200\pi}$$
$$\therefore \log_{10} 100! \approx 157.96964174385386$$

Thus, the number of digits in 100! is 158.

2.3 Inclusion-Exclusion

Reflection 53. This book usually introduces a new topic by way of a problem!

> In a class of 40, many students are collecting the pictures of their favorite rock stars. Eighteen students have a picture of the Beatles, 16 students have a picture of the Rolling Stones and 12 students have a picture of Elvis Presley (this happened a long time ago, when we were young). There are 7 students who have pictures of both the Beatles and the Rolling Stones, 5 students who have pictures of both the Beatles and Elvis Presley, and 3 students who have pictures of both the Rolling Stones and Elvis Presley. Finally, there are 2 students who possess pictures of all three groups.
> Question: How many students in the class have no picture of any of the rock groups?

Reflection 54. This is the grade-school Venn-diagram problem. The answer is 7.

The book also describes one solution. However, it states:

> *We can not find any error in this answer [7], looking at it from any direction. But learning from our previous experience, we must be much more careful: We have to give an exact proof!*
>
> — PG. 30

Reflection 55. And then it demonstrates how to find the answer using the *Inclusion-Exclusion formula.* I was seeing it for the first time, and it felt enigmatic. I also wondered if this formula was designed exclusively to solve problems like this one.

It makes use of exercise 1.3.3: *The number of even subsets of a nonempty n-set is equal to the number of its odd subsets.* However, that observation alone is not enough to set up Inclusion-Exclusion.

I will try to generalize the principle, although I am still looking for my own explanation of how it works. We consider a universe of elements, which is a set, $\mathbb{U}$, and three sets, $\mathbb{A}, \mathbb{B}, \mathbb{C}$.

Let us have n elements in the universe, that is, $|\mathbb{U}| = n$. Every object is a member of either $0, 1, 2$, or 3 of the three sets.

The Inclusion-Exclusion formula then states that

$$n - (n_{\mathbb{A}} + n_{\mathbb{B}} + n_{\mathbb{C}}) + (n_{\mathbb{A}\cap\mathbb{B}} + n_{\mathbb{B}\cap\mathbb{C}} + n_{\mathbb{C}\cap\mathbb{A}}) - n_{\mathbb{A}\cap\mathbb{B}\cap\mathbb{C}} = n_{\mathbb{U}\setminus\mathbb{A}\cup\mathbb{B}\cup\mathbb{C}} \tag{2.5}$$

Actually, I like the following form better:

$$n = n_{\mathbb{U}\setminus\mathbb{A}\cup\mathbb{B}\cup\mathbb{C}} + (n_{\mathbb{A}} + n_{\mathbb{B}} + n_{\mathbb{C}}) - (n_{\mathbb{A}\cap\mathbb{B}} + n_{\mathbb{B}\cap\mathbb{C}} + n_{\mathbb{C}\cap\mathbb{A}}) + n_{\mathbb{A}\cap\mathbb{B}\cap\mathbb{C}} \tag{2.6}$$

If we only had two sets, $\mathbb{A}, \mathbb{B}$, then it takes the form:

$$n = n_{\mathbb{U}\setminus\mathbb{A}\cup\mathbb{B}} + (n_{\mathbb{A}} + n_{\mathbb{B}}) - n_{\mathbb{A}\cap\mathbb{B}} \tag{2.7}$$

The 2-set formula 2.7, when stated in words, is clear: To calculate the number of elements in the universe n, we start with the elements that are not members of any of the sets (that is, they belong to the set $\mathbb{U}$

minus the union of $\mathbb{A}$ and $\mathbb{B}$: $\mathbb{U} \setminus (\mathbb{A} \cup \mathbb{B})$) and add to it the number of elements in $\mathbb{A}$ and the number of elements in $\mathbb{B}$, but in doing so, we add each element common to both the sets twice; therefore we must subtract the number of common elements from the sum, $n_{\mathbb{A}\cap\mathbb{B}}$.

This process can be continued to any finite number of sets. Adopting it to four sets $\mathbb{A}, \mathbb{B}, \mathbb{C}, \mathbb{D}$:

$$\begin{aligned} n = {} & n_{\mathbb{U}\setminus\mathbb{A}\cup\mathbb{B}\cup\mathbb{C}\cup\mathbb{D}} \\ & + (n_{\mathbb{A}} + n_{\mathbb{B}} + n_{\mathbb{C}} + n_{\mathbb{D}}) \\ & - (n_{\mathbb{A}\cap\mathbb{B}} + n_{\mathbb{B}\cap\mathbb{C}} + n_{\mathbb{C}\cap\mathbb{D}} + n_{\mathbb{D}\cap\mathbb{A}}) \\ & + (n_{\mathbb{A}\cap\mathbb{B}\cap\mathbb{C}} + n_{\mathbb{A}\cap\mathbb{B}\cap\mathbb{D}} + n_{\mathbb{B}\cap\mathbb{C}\cap\mathbb{D}}) \\ & - (n_{\mathbb{A}\cap\mathbb{B}\cap\mathbb{C}\cap\mathbb{D}}) \end{aligned} \tag{2.8}$$

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